DEPARTMENT OF DEFENSE (DoD)
SMALL BUSINESS INNOVATION RESEARCH (SBIR)

SBIR X25.5 Annual Program Commercial Solutions Opening (CSO)

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I. INTRODUCTION

a. Program on Innovation Open Topics

15 U.S.C. §638 (ww) requires DOD to establish innovation open topic activities to:

- increase the transition of commercial technology to the DoD;
- expand the small business nontraditional industrial base;
- increase commercialization derived from DoD investments; and
- expand the ability for qualifying SBCs to propose technology solutions to meet DoD needs.

Unlike conventional topics, which specify the desired technical objective and output, open topics use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

Unless otherwise noted in the topic, an SBC may only submit one proposal to each open topic. If an SBC submits more than one proposal for a single open topic, only the most recent certified proposal submitted prior to the submission deadline will receive an evaluation. All previously submitted proposals for the same open topic will be marked nonresponsive and will not receive an evaluation.

b. Topic Release Instructions and Open Topic Solicitation Methodology

DoD Services and Components may elect to release open topics under this CSO once a month throughout fiscal year 2025. New open topics will be published on the first Wednesday of the month, with proposal open and close dates clearly stated within the Topic Release Instructions.

The instructions contained in this CSO are applicable to all subsequent releases. Additional instructions specific to an individual Topic Release will be contained within the Topic Release Instructions. Each release will have corresponding open and close dates applicable only to those topics contained within the release. **The DoD will accept proposals only in response to specified topic calls.**

II. PROGRAM DESCRIPTION

a. Three-Phased Program

The SBIR Program is a three-phase program. The purpose of **Phase I** is to determine, to the extent possible, the scientific, technical, and commercial merit and feasibility of ideas submitted under the SBIR Program. Phase I awards are made in accordance with the SBA Policy Directive guidelines, current version. The period of performance is *generally* between six to twelve months with twelve months being the maximum period allowable. Proposals should concentrate on research or research and development which will significantly contribute to proving the scientific and technical feasibility, and commercialization potential of the proposed effort, the successful completion of which is a prerequisite for further DoD support in Phase II. Proposing small business concerns are encouraged to consider whether the research or research and development being proposed to DoD Components also has private sector potential, either for the proposed application or as a base for other applications.

Phase II awards will be made to proposing small business concerns (SBCs) based on the results of their Phase I effort and/or the scientific merit, technical merit, or commercialization potential of a Phase II proposal. Phase II awards are made in accordance with the SBA Policy Directive guidelines, current version. The period of performance is *generally* 24 months. Phase II is the principal research or research and development effort and is expected to produce a well-defined deliverable prototype. A Phase II contractor may receive up to one additional, sequential Phase II award for continued work on the project.

Under 15 U.S.C. §638(cc), as amended by NDAA FY12, Sec. 5106, and further amended by NDAA FY19, Sec. 854, PILOT TO ALLOW PHASE FLEXIBILITY, the DoD may issue Phase II SBIR awards to SBCs that did not receive a Phase I SBIR/STTR award for the R/R&D proposed to the agency for funding. Such awards are referred to as Direct-to-Phase-II (D2P2) awards. The DoD makes D2P2 awards by means of clearly-designated D2P2 solicitation topics. These topics accept only D2P2 proposals. These topics result only in D2P2 awards. **The DoD does not accept Phase I proposals for D2P2 topics, and does not accept D2P2 proposals for Phase I topics.**

Under **Phase III**, the proposing SBC is required to obtain funding from either the private sector, a non-SBIR Government source, or both, to develop the prototype into a viable product or non-R&D service for sale in military or private sector markets. SBIR Phase III refers to work that derives from, extends, or completes an effort made under prior SBIR funding agreements, but is funded by sources other than the SBIR Program. Phase III work is typically oriented towards commercialization of SBIR research or technology.

b. Eligibility and Performance Requirements

i. Applicant Must Qualify as a Small Business Concern

Each proposing small business concern must qualify as a small business concern as defined by 13 C.F.R §§ 701-705 at time of award and certify to this in the Cover Sheet (Volume I) section of the proposal. The eligibility requirements for the SBIR/STTR programs are unique and do not correspond to those of other small business programs. Proposing small business concerns must meet eligibility requirements for Small Business Ownership and Control (see 13 CFR § 121.702)

ii. Performance of Work Requirements

A minimum of two-thirds (67%) of the research and/or analytical work in Phase I must be conducted by the proposing Small Business Concern. For Phase II, a minimum of one-half (50%) of the research and/or analytical work must be performed by the proposing small business concern. The percentage of work is measured by both direct and indirect costs. Occasionally, deviations from these SBIR requirements may occur, and must be approved in writing by the Funding Agreement officer after consultation with the agency SBIR/STTR program manager/coordinator.

Note About Proposal Submission: The amounts listed in the Percentage of Work (POW) certification question on the proposal Cover Sheet (Volume I) are derived from SBC-entered information in the Cost Volume (Volume III). Details on the calculation can be viewed in DSIP during proposal submission. If the POW calculations fall below eligibility requirements, the applicant must request an explanatory letter from the DOD SBIR/STTR Contracting officer in order to complete the submission. Approval to **submit** a proposal that deviates from the prescribed POW requirements and Funding Agreement Officer **approval of the proposed deviation** are distinct approvals.

iii. Primary Employment of the Principal Investigator

For both Phase I and II, the primary employment of the principal investigator must be with the proposing small business concern at the time of the award and during the conduct of the proposed effort. Primary employment means that more than one-half of the principal investigator's time is spent with the small business. Primary employment with a small business concern precludes full-time employment at another organization.

iv. Location of the Work

For both Phase I and Phase II, all research or research and development work must be performed by the small business concern and its subcontractors in the United States. Based on a **rare and unique circumstance**, the DOD may approve a particular portion of the R/R&D work to be performed or

obtained in a country outside of the United States. For example, if a supply or material or other item or project requirement is not available in the United States.

c. Ownership by Multiple Venture Capital Operating Companies (VCOCs), Hedge Funds, and Private Equity Firms

Proposing small business concerns that are owned in majority part by multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds are eligible to submit applications and receive awards under this CSO. Any proposing small business concern that is owned, in whole or in part, by any VCOC, hedge fund, and/or private equity fund must identify each foreign national, foreign entity, foreign person, or foreign government holding or controlling greater than a 5% equity stake in the proposing small business concern, whether such equity stake is directly or indirectly held. The proposing small business concern must also identify any and all of its ultimate parent owner(s) and any other entities and/or individuals owning more than a 5% equity stake in its chain of ownership.

See Topic Release Instructions evaluation criteria for additional information regarding SBCs owned in majority part by multiple VCOCs, hedge funds, or private equity firms.

d. Performance Benchmark Requirements/Increased Minimum Performance Standards for Experienced Firms

Applicants with multiple prior SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The Phase I to Phase II Transition Rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The Commercialization Benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' application for more experienced firms. Find detailed information on benchmark calculations, increased performance standards for more experienced firms and consequence of failure to meet benchmarks here. SBA will notify companies failing either benchmark and the relevant officials at the participating agencies.

The SBIR/STTR Policy Directive defines the Departments of the Army, Navy, and Air Force each as its own federal agency, and the remaining DoD Components as an executive agency of the Department of Defense. Therefore, companies that fail to meet either of the benchmarks under the Increased Performance Standards for More Experienced Firms may not receive more than an overall total of 80 awards from DoD, as detailed in the breakdown below:

Army – 20 total Phase I and Direct to Phase II awards
Navy – 20 total Phase I and Direct to Phase II awards
Air Force – 20 total Phase I and Direct to Phase II awards
All other DoD Components combined – 20 total Phase I and Direct to Phase II awards

In addition, SBA has posted a Guide to SBIR/STTR Program Eligibility to help small businesses understand program eligibility requirements, determine if they will be eligible at award, and accurately complete necessary certifications. Note, applicants are required to certify "Yes", "No", or "N/A" in DSIP Firm Certifications. Applicants who indicate that they are not eligible to participate may be disqualified.

Companies' benchmark information is not available to the public.

e. Direct to Phase II (D2P2) Proposal Standards

Awards under clearly-designated D2P2 topics will provide funds to conduct further R/R&D for non-defense commercial solutions to meet specific DOD end-users' and customers' needs. These D2P2

awards are intended to provide a commercialization path, and are not the proposed solutions' final step. Second phase commitments from non-SBIR Federal and non-Federal sources are not mandatory. Cost-share is also not required. However, second phase commitments may serve as commercial potential evidence in accordance with 15 USC 638(e)(4)(B)(ii). Applicants must adequately document completion of the Phase I feasibility requirement*.

Phase II proposals require a comprehensive, detailed effort description. Proposals should demonstrate sufficient technical progress or problem-solving results to warrant more extensive R/R&D. Developing technologies with commercial and military potential is extremely important. Particularly, DOD is seeking proposals emphasizing technologies' dual-use applications and commercialization.

*NOTE: The applicant shall provide information to enable the agency to make the 15 U.S.C. 638(cc) determination of scientific and technical feasibility and merit. Applicants are required to provide information demonstrating scientific and technical merit and feasibility has been established as part of the Technical Volume (Volume II). **Feasibility documentation cannot be based upon or logically extend from any prior or ongoing federally funded SBIR or STTR work.**

f. Discretionary Technical and Business Assistance (TABA)

The DOD does not offer TABA. Applicants shall not propose costs associated with TABA and such costs if proposed will not be accepted by the DOD.

g. State and Other Available Assistance

Many states have established programs to provide services to those proposing SBCs and individuals wishing to participate in the Federal SBIR Program. These services vary from state to state, but may include:

- Information and technical assistance;
- Matching funds to SBIR recipients;
- Assistance in obtaining Phase III funding.

SBCs may seek general administrative guidance from small and disadvantaged business utilization specialists located in various defense contract management activities throughout the continental United States.

h. Fraud and Fraud Reporting

Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. Sec 1001), punishable by a fine of up to $10,000, up to five years in prison, or both. The DoD Office of Inspector General Hotline ("Defense Hotline") is an important avenue for reporting fraud, waste, abuse, and mismanagement within the DoD. The Office of Inspector General operates this hotline to receive and investigate complaints or information from contractor employees, DoD civilians, Service members, and the public. Individuals who wish to report fraud, waste or abuse may contact the Defense Hotline at (800) 424-9098 between 8:00 a.m. and 5:00 p.m. Eastern Time or visit their website to submit a complaint. Mailed correspondence should be addressed to the Defense Hotline, The Pentagon, Washington, DC 20301-1900, or email addressed to hotline@dodig.mil.

III. PROPOSAL SUBMISSION

a. DSIP Submission

DSIP is the official portal for DoD SBIR/STTR proposal submission and its use is mandatory for all applicants and Government users. Applicants are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed guidance and Job Aids on DSIP proposal

submission are found here. Applicants are required to submit proposal documents into DSIP within the file size limitations indicated in DSIP.

b. Effect of Deadlines

Deadline for Receipt: Complete proposals must be certified and submitted in DSIP no later than the close date of the release listed on the Topic Release Instructions. Proposals cannot be submitted in the DSIP after the deadline is reached and will not be accepted or evaluated. No exceptions will be made to the submission deadline. DoD recommends early submission, as computer traffic intensifies near announcement closing. Do not wait until the end of the submission period to submit proposal materials. DOD is not responsible for missed proposal submission due to system lag or inaccessibility.

c. Proposal Volumes

The final proposal submission includes successful completion of all firm level forms, all required proposal volumes, and electronic corporate official certification. Firm-level forms include:

- Firm Certifications
- Audit Information
- Company Commercialization Report (which then becomes Volume 4)

Although signatures are not required on the electronic forms at the time of submission, the proposal must be certified electronically by the corporate official for it to be considered submitted. If the proposal is selected for negotiation and possible award, the DOD program will contact the applicant for signatures prior to award. **Please plan to submit proposals as early as possible to allow time for troubleshooting any possible issues before the CSO close. DSIP Support is unable to assist with submission issues once a deadline has passed and cannot provide submission extensions. DoD is not responsible for missed proposal submission due to system latency.**

i. Proposal Cover Sheet (Volume 1)

The proposal Cover Sheet (Volume 1) is prepared on DSIP. The Cover Sheet (Volume 1) must include a brief technical abstract that describes the proposed R&D project and an anticipated benefits and potential commercial applications discussion. Each section should be no more than 3,000 characters.

Do not include proprietary or classified information in the proposal Cover Sheet (Volume 1). If your proposal is selected for negotiation and possible award, the technical abstract and anticipated benefits discussion may be publicly released online. DSIP will assign a proposal number once the Cover Sheet (Volume 1) is saved. Applicants may modify the Cover Sheet (Volume 1) as needed until the Topic Release closes.

NOTE: the amounts listed in the percentage of work (POW) certification question on the proposal cover sheet are derived from SBC-entered information in the cost volume (Volume 3). Details on the calculation can be viewed in DSIP during proposal submission.

ii. Technical Volume Format & Content (Volume 2)

The technical volume format and content instructions will be addressed in the Topic Release Instructions.

iii. Cost Volume (Volume 3)

The cost volume format and instructions will be addressed in the Topic Release Instructions.

iv. Company Commercialization Report (Volume 4)

The Company Commercialization Report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. SBIR and STTR awardees are required by SBA to update and maintain their organization's CCR on SBIR.gov. Commercialization information is required upon

completion of the last deliverable under the funding agreement. Thereafter, SBIR and STTR awardees are requested to voluntarily update the information in the database annually for a minimum period of five years.

If the proposing firm has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, regardless of whether the project has any commercialization to date, a PDF of the CCR must be downloaded from SBIR.gov and uploaded to the Firm Forms section of DSIP by the Firm Admin. Firm Forms are completed by the DSIP Firm Admin and are applied across all proposals the firm submits. The DSIP CCR requirement is fulfilled by completing the following:

1. Log into the firm account at https://www.sbir.gov/.
2. Navigate to My Dashboard > My Documents to view or print the information currently contained in the Company Registry Commercialization Report.
3. Create or update the commercialization record, from the company dashboard, by scrolling to the "My Commercialization" section, and clicking the create/update Commercialization tab under "Current Report Version". Please refer to the "Instructions" and "Guide" documents contained in this section of the Dashboard for more detail on completing and updating the CCR. Ensure the report is certified and submitted.
4. Click the "Company Commercialization Report" PDF under the My Documents section of the dashboard to download a PDF of the CCR.
5. Upload the PDF of the CCR (downloaded from SBIR.gov in the previous step) to the Company Commercialization Report in the Firm Forms section of DSIP. This upload action must be completed by the Firm Admin.

In Volume 4 of the DSIP proposal submission, the applicant will be prompted to answer: "Do you have a new or revised Company Commercialization Report to upload?" There are three possible courses of action:

a. If the applicant has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards and **has a new or revised CCR from SBIR.gov to upload to DSIP**, select YES.
   1. If the user is the firm admin, they can upload the CCR PDF from SBIR.gov directly on this page. It will also be updated in the "Firm Forms" and be associated with all new or in-progress proposals the applicant submitted. If the user is not the firm admin, they will receive a message that they do not have access and must contact the firm admin to complete this action.
   2. **WARNING:** Uploading a new CCR under the DSIP "Firm Forms" section or clicking "Save" or "Submit" in one proposal's Volume 4 is considered a change for ALL proposals under any open BAAs or CSOs. If an applicant has previously certified and submitted any Phase I or Direct to Phase II proposals under ***any*** BAA or CSO ***still open***, those proposals will be automatically reopened. Applicant will have to recertify and resubmit affected proposals. If an applicant does not recertify or resubmit affected proposals, they will not be considered fully submitted and will not be evaluated.

b. If the applicant has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, and **no new or revised CCR from SBIR.gov to upload to DSIP**, select NO.
   1. If a prior CCR was uploaded to the "Firm Forms", the applicant will see a file dialog box at the bottom of the page and can view the previously uploaded CCR. This read-only access allows the applicant to confirm the firm admin uploaded the CCR.
   2. If no file dialog box appears at the bottom of the page **there is no previously uploaded CCR in the DSIP "Firm Forms**." To fulfill the DSIP CCR requirement the firm admin

must follow steps 1-5 listed above to download a CCR PDF from SBIR.gov and upload it to the DSIP "Firm Forms" to be included with all proposal submissions.

c. If the applicant has **NO** prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, the CCR upload from SBIR.gov is not required and SBC will select NO. The proposal's CCR section will be marked complete.

Please refer to the Topic Release Instructions for details on how the CCR information will be considered during proposal evaluations.

v. Supporting Documents (Volume 5)

Volume 5 is provided for applicants to submit additional documentation to support the Cover Sheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3).

The following documents may be included in Volume 5, if applicable to the proposal. Refer to Topic Release Instructions for additional Volume 5 requirements.

Reminder: A completed proposal submission in DSIP does NOT indicate the mandatory supporting documents have been uploaded in accordance with the Topic Release Instructions.

- Letters of support
- Additional cost information
- Funding agreement certification
- Technical data rights (assertions)
- Lifecycle certification
- Allocation of rights
- Verification of Eligibility of Small Business Joint Ventures, if applicable
- DD Form 2345, Military Critical Technical Data Agreement, if applicable

vi. Fraud, Waste, and Abuse (FWA) Training (Volume 6)

FWA Training is required for DoD SBIR/STTR proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA. The training currently consists of a 3-page PDF, consistent with the tutorial provided by the SBA. This training material must be thoroughly reviewed once per year and can be found here and in the DSIP proposal submission module for Volume 6. Plan time to review the tutorial during completion of Volume 6, prior to the proposal submission deadline. The individual designated as the Proposal Owner must complete this training on behalf of the applicant small business.

vii. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)

In accordance with 15 U.S.C. §638 (vv) and the SBA SBIR/STTR Policy Directive, the DoD will review all proposals submitted in response to this CSO to assess security risks of SBCs seeking a federally funded award. SBCs must complete the DSIP Volume 6 webform "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (NOTE: PDF uploads are no longer accepted). The corporate official cannot certify and submit the full proposal until the Volume 6webform is fully completed and submitted.

Please be aware that the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" form WILL NOT be accepted as a supporting document in DSIP's Volume 5 proposal submission. Do not upload any previous versions of this form to Volume 5.

d. DSIP Support

DSIP Support is unable to assist with submission issues once a deadline has passed and cannot provide submission extensions. If the proposal status is "In Progress" or "Ready to Certify" it will NOT be considered submitted, even if all volumes are added prior to the CSO close date. The proposing SBC may modify all proposal volumes prior to the Topic Release Instructions closing date.

The proposing SBC may modify all proposal volumes prior to the CSO closing date, however doing so will un-certify any previously certified and submitted proposal documents. Edited proposals will revert "In Progress" or "Ready to Certify" status and will not be considered submitted, even if all volumes are added prior to the Topic Release Instructions closing date.

DSIP Support Desk: Refer to the Customer Support Document at https://www.dodsbirsttr.mil/submissions/api/public/download?uploadId=MTUzODQ1OA==&fileName=DSIP_Customer_Support_Document.pdf&showOnWeb=true for general information regarding the DoD SBIR/STTR process. For additional assistance with the DSIP application, please visit https://www.dodsbirsttr.mil/submissions/learning-support/.

Email DSIP Support at DoDSBIRSupport@reisystems.com only for further assistance with issues pertaining directly to the DSIP application. Questions submitted to DSIP Support will be addressed in the order received during normal operating hours (Monday through Friday, 9:00 a.m. to 5:00 p.m. ET).

Applicants are responsible for following ALL troubleshooting guidance provided by DSIP Support. Applicants are responsible for reengaging with DSIP Support if provided troubleshooting tips do not resolve applicant proposal submission issues. Applicant proposal submission failure is rarely, if ever, attributable to DSIP performance issues.

e. Export-Controlled Technologies

DD Form 2345 - ITAR/EAR Certification: For proposals whose solutions fall under export- control regulations (either International Traffic in Arms (ITAR) or Export Administration Regulations (EAR)), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website, http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP/DD 2345Instructions.aspx. Approval of the DD Form 2345 will be verified if the proposal is chosen for award.

IAW DFARS 225.7901-4, Export Control Contract Clauses, the clause found at DFARS 252.225- 7048, Export-Controlled Items (June 2013), must be included in all solicitations and contracts; therefore, all awards resulting from this CSO will include DFARS 252.225-7048 or similar verbiage.

Note: for proposals whose solutions fall under export-control regulations, the Principal Investigator must be a U.S. citizen or lawful permanent resident.

f. Classified Proposals

Applicants may **not** submit classified proposals in response to this CSO. If efforts will require classified work during Phase II performance, the applicant must have or obtain a facility clearance.

Additional information on facility and personnel clearance procedures and requirements can be found at the Defense Counterintelligence and Security Agency site, http://www.dss.mil/index.html.

g. Promotional Materials

Promotional and non-project related proposal content is discouraged. Additional information provided via website or other communication medium, except where explicitly requested, will not be considered.

h. Prior, Current, or Pending Support of Similar Proposals or Awards

While it is permissible, with proposal notification, to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program solicitations, it is unlawful to enter negotiation for contracts requiring essentially equivalent effort. If there is any question concerning prior, current, or pending support of similar proposals or awards, it must be disclosed as early as possible and declared on the proposal Cover Sheet (Volume 1).

i. Technical Baseline

Department of the Air Force ONLY: As RDT&E programs, SBIR/STTR efforts are subject to scientific and technical literature searches in order to establish technical baseline, avoid duplication, and justify funding IAW AFI 61-201, Management of Scientific and Technical Information (STINFO) (29 January 2019) and AFRL/CA guidance, Streamlined and Waivered SBIR/STTR Specified Requirements (05 Aug 2019).

j. Treatment of Commercial Off-the-Shelf (COTS) and Non-Developmental Items (NDIs)

COTS/NDIs: Items purchased directly from a commercial source that can be utilized without alteration through design and development, or without modification, are classified as COTS or NDI. This includes, for example, ready-to-use products, training services, and software licenses for ready-to-use software to satisfy service needs (including Software as a Service (SaaS)).

Purchases of COTS and NDIs for use, including the first article and associated first article acceptance testing and related minor adjustments are not suitable for SBIR/STTR funding. The acquisition of commercial items or NDIs for testing and operational evaluation that do not require R/R&D engineering, design or integration effort will not be funded by SBIR/STTR funds. However, if an end item requires design and development in order to accept the COTS or NDI, funding for design and development effort could be funded by SBIR/STTR funds. If a COTS or NDI is required for RDT&E test purposes, the cost could be funded by SBIR/STTR funds.

Modified COTS/Modified NDIs: Commercially available items that must be modified to satisfy user requirements are classified as "modified COTS" or "modified NDI" articles. In this instance, the first article, modification of the first article, and first article testing could be funded by SBIR/STTR funds. Follow-on purchases will not be funded by SBIR/STTR funds. The number of “modified” first articles bought with SBIR/STTR funds will not exceed the quantity needed to conduct the RDT&E acceptance tests. Modification is technology refreshment significantly changing the end item’s performance envelope. If the commercially available item is modified and requires testing prior to approval for service use or inventory, it may be funded by SBIR/STTR funds, as all developmental items. In contrast to modification, continuous technology refreshment is the intentional insertion of newer technology to improve reliability, improve maintainability, reduce cost, and/or add minor performance enhancement, typically in conjunction with depot or field level maintenance. The insertion of such technology into end items as part of maintenance is not funded by SBIR/STTR funds.

k. Use of Commercial Item Procedures

The Government may acquire innovative commercial items, technologies, or services using a competitive procedure called a Commercial Solutions Opening (CSO) using DFARS Subpart 212.70. Under a CSO, DoD may competitively select proposals received in response to a general solicitation, similar to a Broad Agency Announcement, based on proposal review by Government scientific, technological, or other subject-matter experts.

CSO use in accordance with DFARS Subpart 212.70 is considered competitive for the purposes of 10 U.S.C. chapter 127 and FAR 6.102. The DOD will treat items, technologies, and services acquired under this CSO as commercial items. The full definition of commercial items, i.e., products and services, under FAR 2.101 applies. This CSO includes no requirement for a commercial product or service to be an already-developed, off-the-shelf item.

l. Marking Proprietary Proposal Information

The DOD assumes no liability for disclosure or use of unmarked data and may use or disclose such data for any purpose. Restrictive notices notwithstanding, proposals and final reports submitted through DSIP may be handled, for administrative purposes only, by support contractors. All support contractors are bound by appropriate non-disclosure agreements.

Information contained in unsuccessful proposals will remain the property of the Applicant. The Federal Government may, however, retain copies of all proposals. Public release of information in any proposal submitted will be subject to existing statutory and regulatory requirements. If proprietary information is provided by an Applicant in a proposal, which constitutes a trade secret, commercial or financial information, it will be treated in confidence, to the extent permitted by law, provided that the proposal is clearly marked by the applicant.

IV. CERTIFICATIONS AND REGISTRATIONS

a. System for Award Management (SAM) Registration

Before the DOD can award a contract, applicant firms must be registered in the System for Award Management (SAM). SAM allows firms interested in conducting business with the federal government to provide basic information on business structure and capabilities as well as financial and payment information. To register, visit http://www.sam.gov. Firms should login to SAM and ensure the firm's registration is active and representations and certifications are up-to-date to avoid delay in award.

Registration in SAM will generate the Unique Entity ID (UEI) number and the Commercial and Government Entry (CAGE) code. The UEI is required for registration in the U.S. Small Business Administration's (SBA) Company Registry.

SAM.gov registration should reflect ALL AWARDS for Purpose of Registration. Firms registered to receive grants only will be ineligible.

b. SBA Company Registry

Proposing SBCs must be registered in the SBA Company Registry. SBCs will be required to verify registration by providing the SBC Control ID and Proof of Registration/Certification during proposal submission.

c. Defense SBIR/STTR Innovation Portal (DSIP) Registration

Individuals from proposing SBCs must be registered in the DSIP to prepare and submit proposals. Proposing SBCs submitting through this site for the first time will be asked to register. All users are required to have an individual user account to access DSIP. It is recommended proposing SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process.

DSIP user accounts are authenticated by Login.gov. Users who do not already have a Login.gov account will be required to create one. Users who already have a Login.gov account can link their existing

Login.gov account with their DSIP account. Job Aids and Help Videos to walk you through the process are in the Learning & Support section of DSIP.

Be advised the sharing of accounts and passwords is a violation of the Terms of Use for Login.gov and DoD policy.

Registered SBCs will have a designated DSIP Firm Admin responsible for creating the Firm PIN, controlling access for other users in the SBC and completing and maintaining the Firm-level forms, which must be completed before any proposals can be submitted.

Users should complete their account registrations as soon as possible to avoid any delays in proposal submissions.

NOTE: The DSIP application is only accessible from within the United States, which is defined as the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

Email DSIP Support at DoDSBIRSupport@reisystems.com **only** for assistance with using the DSIP application. Questions regarding DSIP can be emailed to DSIP Support and will be addressed in the order received, during normal operating hours (Monday through Friday, 9:00 a.m. to 5:00 p.m. ET). Please include information on your firm, a proposal number (if applicable), and screenshots of any pertinent errors or issues encountered.

DSIP Support cannot provide updates to proposal status after submission, such as proposal selection/non-selection status or contract award status. Do not attempt to directly submit proposal documents to the above email address. Such transmissions do not constitute proposal submission and such materials will be discarded.

d. Required Certifications

At the time of proposal submission, each SBC must certify via the Cover Sheet (Volume 1) of the proposal that it meets the size, ownership, and other requirements of the SBIR Program. In addition, the Policy Directive includes certification requirements set forth in the SBIR and STTR Extension Act of 2022 (Public Law 117-183). SBCs are required to certify that they are meeting the Program's requirements during the life cycle of the funding agreement: at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the total Phase II award amount, and prior to final payment on the Phase II award.

e. Due Diligence Program to Assess Security Risks

15 U.S.C. §638 (vv) requires the DoD, in coordination with the SBA, to establish and implement a due diligence program to assess and, when possible, mitigate national security risks from SBCs seeking a federally funded award. The Department will use the proposal and information in response to the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (proposal submission Volume 7) to conduct a risk-based due diligence review of the following areas: cybersecurity practices; patent analysis; employee analysis and foreign ownership, including the SBC's financial ties and obligations (which shall include surety, equity, and debt obligations); and SBC employees' ties to a foreign country, foreign person, or foreign entity. The Department will also assess proposals utilizing open-source analysis and analytical tools, for the purpose of confirming the accuracy of the information provided as well as determining if the proposing SBC failed to disclose the information set forth in 15 U.S.C. 638(g)(13).

After reviewing the proposing SBC's responses to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries, if DoD determines it appropriate the Department may ask the SBC to provide true copies of any contractual or financial obligation or other agreement specific to a business arrangement or joint-venture like arrangement with an enterprise owned by a foreign state or any foreign entity in effect during the five-year period preceding the SBC's proposal submission.

The DoD may not make awards that pose an unacceptable risk to national security. If the risk-based due diligence review concludes that an SBC should not be eligible for the specific SBIR or STTR award due to a national security risk that cannot be adequately mitigated, the proposal will not receive consideration for possible award regardless of the results of the technical review of the proposal. Attachment 2: Defense SBIR and STTR Due Diligence Program Common Risk Matrix of the memo titled "Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program", dated 13 May 2024, provides details on the factors for assessing SBC risk during the due diligence review.

The DoD has partnered with Project Spectrum to provide an online course on Understanding Foreign Ownership, Control, or Influence (FOCI). This course defines FOCI, explains what it means to be under FOCI, and details FOCI's effect on a company seeking initial or continued eligibility for access to a federally funded award. Small business concerns can register and access this course by following the instructions below:

- Go to projectspectrum.io
- Click "Profile/Dashboard in the top right and then click "Sign Up" from the dropdown menu.
- Follow the instructions to sign up for an account. Descriptions of the account types are provided below each option.
- Verify your email by entering the code sent to the email address you provided when signing up.
- Log in to Project Spectrum by clicking "Profile/Dashboard > Login" in the top right.
- Find the Training Course on "Understanding Foreign Ownership, Control, or Influence (FOCI)" by clicking "Courses > Training Courses"
- Copy the provided password.
- Click on the course and log in to Encite.io using your email address and the copied password.
- Enroll in the course and click "Enter" to begin.

For assistance with registration or access to the Project Spectrum website, please contact support@projectspectrum.io.

f. Joint Ventures

Joint ventures and limited partnerships are eligible to submit proposals to this solicitation, provided that the entity created qualifies as a small business IAW the Small Business Act, 13 U.S.C. § 121.701. Proposing small business concerns must disclose joint ventures with existing (or planned) relationships/partnerships with any foreign entity or foreign government-controlled companies.

Joint venture applicants must submit the representations found here by uploading the representations to Volume 5, Supporting Documents.

g. Conflicts of Interest

Contract awards to an SBC owned by or employing current, or previous, Federal Government employees could create conflicts of interest for those employees, which may be a violation of federal law.

In accordance with FAR 9.5, proposing SBCs are required to identify and disclose all facts relevant to potential organizational conflicts of interest (OCIs) involving the proposing SBC's organization and any

proposed team member (sub-awardee, consultant). The proposing SBC is responsible for providing this disclosure with each submitted proposal. The disclosure must include the proposing SBC's, and as applicable, proposed team member's OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposing SBC has taken, or intends to take, to prevent the existence of conflicting roles that might bias the proposing SBC's judgment, and to prevent the proposing SBC from having unfair competitive advantage. The OCI mitigation plan will specifically discuss the disclosed OCI in the context of each of the OCI limitations outlined in FAR 9.505-1 through FAR 9.505-4.

As part of the FAR 9.5 disclosure requirement, a proposing SBC must affirm whether the proposing SBC or any proposed team member (sub-awardee, consultant) is providing SETA, A&AS, or similar support to any DoD Service/Component office(s) under: (a) a current award or sub-award; or (b) a past award or sub-award that ended within one calendar year prior to the proposal's submission date. If SETA, A&AS, or similar support is or was provided to any DoD Service/Component office(s), the proposal must include:

- The name of the DoD Service/Component office receiving the support;
- The prime contract number;
- Identification of proposed team member (sub-awardee, consultant) providing the support; and
- An OCI mitigation plan in accordance with FAR 9.5.

Government Procedures

In accordance with FAR 9.503, 9.504 and 9.506, the Government will evaluate OCI mitigation plans to avoid, neutralize, or mitigate potential OCI issues before award and determine whether it is in the government's interest to grant a waiver. The government may require proposing SBCs provide additional information to support evaluation of the proposing SBC's OCI mitigation plan. If the government determines a applicant failed to fully disclose an OCI; or failed to provide a government waiver as described above; or failed to reasonably provide additional information the government requested when evaluating the applicant's OCI mitigation plan, the government may reject the proposal and withdraw it from consideration for award.

h. Research Involving Human Subjects/Human Subject Research (RIHS/RHS)

All research involving human subjects, to include use of human biological specimens and human data, shall comply with the applicable federal and state laws, and agency policy/guidelines for human subject protection.

Institutions receiving funding for research involving human subjects must provide documentation of a current federal assurance of compliance with federal regulations for human subject protection; for example a Department of Health and Human Services, Office for Human Research Protections Federalwide assurance (http://www.hhs.gov/ohrp). The DOD may also request additional federal assurance documentation. All institutions engaged in human subject research, to include subcontractors, must also have a valid assurance. In addition, personnel involved in human subjects research must provide documentation of completed appropriate training for the protection of human subjects. Institutions proposing to conduct human subject research that meets one of the exemption criteria in 32 CFR 219.101 are not required to have a federal assurance of compliance. Proposing SBCs should clearly segregate research activities involving human subjects from other R&D activities in their proposal. If selected, institutions must also provide documentation of institutional review board (IRB) approval, or a determination from an appropriate official in the institution, that the work meets one of the exemption criteria with 32 CFR 219. As part of the IRB review process, evidence that all investigators are appropriately trained should accompany the protocol. The protocol, separate from the proposal, must include a detailed description of the research plan, study population, risks and benefits of study participation, recruitment and consent process, and data collection and analysis.

The amount of time required for the IRB to review and approve the protocol will vary based on the IRB's procedures, the complexity of the research, the level of risk to study participants and the responsiveness of the investigator. The average IRB approval process can last between one and three months. Once the IRB has approved the research, the awarding DOD will review the protocol and the IRB's determination to ensure that the research will be conducted in compliance with both DoD and DOD policies. The DOD review process can last between three to six months. Ample time should be allotted to complete both the IRB and DOD approval processes prior to recruiting subjects. No funding can be used towards human subject research until ALL approvals are granted. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.

i. Research Involving Recombinant DNA Molecules

All research involving recombinant DNA molecules shall comply with the applicable federal and state law, regulation, and additional agency guidance. An institutional biosafety committee must approve the research. In addition to the standard federal and DoD procurement certifications, the SBA SBIR Policy Directive requires that applicants provide certain information at time of award and during the award life cycle. Each proposing SBC must provide this additional information at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the Phase II total award amount, and prior to final payment on the Phase II award.

j. Use of Unmanned Aircraft Systems (UAS)

All activities involving the operation of Unmanned Aerial Systems shall be conducted IAW Department of Defense Memo entitled "Guidance for the Domestic Use of Unmanned Aircraft Systems in the U.S. National Airspace", dated 18 August 2018 and DOD memo entitled, "Delegation Authority to Approve Exemptions for Using Commercial-Off-The-Shelf Unmanned Aerial Systems in Support of Urgent Needs", dated 16 November 2018. Contractor personnel operating DoD or DoD- contracted UAS in U.S. National Airspace, whether or not the domestic use is related to an intelligence activity, will comply with all laws, regulations, and guidance relating to the operation of UAS in U.S. National Airspace. Specific programs utilizing UAS are required to obtain an exemption for operational use of a UAS; therefore, additional documentation may be requested by the awarding agency to support an exemption request. The DOD exemption approval process can require as long as two months. **No funding can be used for contracted-UAS operations until ALL approvals are granted.**

k. Federal Acquisition Supply Chain Security Act Orders

FAR 52.204-29 Federal Acquisition Supply Chain Security Act (FASCA) Orders—Representation and Disclosures and FAR 52.204-30 FASCA Orders—Prohibition are included in this solicitation. In accordance with FAR 52.204-29 and FAR 52.204-30, proposing SBCs must review FASCSA orders here for covered articles, or any products or services produced or provided by a source, that an applicable FASCSA order prohibits.

During contract performance, the contractor shall review SAM.gov at least once every three months, or as the contracting officer advises, to check for covered articles, or products or services produced subject as part of any new FASCSA order(s) that could impact their supply chain, and report to the contracting officer any covered article, or product or service produced or provided by a source provided to the government or used during the contract performance.

The proposing SBC represents that, via proposal submission under this CSO, it conducted a reasonable inquiry, and it does not propose to provide or use any covered article, or any products or services produced or provided by a source, if an applicable FASCSA prohibited the covered article or the source effective this CSO's issue date.

V. MISCELLANEOUS INFORMATION

a. Award Information

Awards are executed as firm-fixed-price contracts or Other Transactions Authority (OTA) under 10 USC 4021 (Research Projects), 10 USC 4022 (Prototype), or 10 USC 4023 (Procurement for Experimental Purposes). Follow-on production OTs are possible as a result of OT awards made under this solicitation.

b. Fee or Profit

The DOD is required to provide reasonable fee or profit, consistent with normal profit margins provided to profit-making firms for R/R&D work. As such, the DOD reserves the right to reject any proposal that fails to account for reasonable profit or fee.

c. Payment Milestones

The frequency of payments under awards resulting from this CSO will be determined by the agreed-upon milestone schedules incorporated into the award document. Technical Milestone schedules are used as a means to monitor technical progress, to mitigate technical and cost risk, and to address the cashflow needs of awardees.

d. Contractual Requirements

Awardees will be required to make certain legal commitments through acceptance of Government contract clauses or OT articles. The following outline is illustrative of provisions required by the FAR or statute for inclusion in Phase II awards. This is not a complete list, nor does it contain specific clause/article wording.

i. Standards of Work

Work performed under the award must conform to professional standards.

ii. Inspection

Work performed under the award is subject to Government inspection and evaluation at all reasonable times.

iii. Examination of Records

The Comptroller General (or a fully authorized representative) shall have the right to examine any directly pertinent awardee records involving award transactions.

iv. Termination For Cause

The Government may terminate the award for failure to perform the negotiated effort.

v. Termination for Convenience

The Government may terminate the award at any time if it deems termination in its best interest. The firm will be compensated for work performed and reasonable termination costs.

vi. Disputes

Any award dispute not resolved by agreement shall be decided by the Contracting or Agreements Officer with right of appeal.

vii. Contract/OT Work Hours

The awardee may not require an employee to work more than eight hours a day or forty hours a week unless compensated accordingly through overtime pay.

viii. Equal Opportunity

The firm will not discriminate against any employee or employee applicant because of race, color, religion, sex, or national origin.

ix. Affirmative Action for Veterans

The awardee will not discriminate against any employee or applicant for employment because he or she is a disabled veteran.

x. Affirmative Action for Handicapped Individuals

The awardee will not discriminate against any employee or applicant for employment because he or she is physically or mentally handicapped.

xi. Officials Not to Benefit

No Federal Government official may benefit personally from the SBIR/STTR funding agreement.

xii. Covenant Against Contingent Fees

No person or agency has been employed to solicit or secure the award upon an understanding for compensation except bona fide employees or commercial agencies maintained by the awardee for the purpose of securing business.

xiii. Gratuities

The Government may terminate the award if gratuities have been offered to any Government representative to secure the award.

xiv. Patent Infringement

The awardee shall report each notice or patent infringement claim based on contract performance.

xv. Military Security Requirements

The awardee shall safeguard classified information associated with the contracted work IAW applicable regulations.

xvi. American-Made Equipment and Products

The awardee will purchase equipment and/or products in support of the SBIR award from U.S.-based sources whenever possible.

xvii. Unique Identification (UID)

If hardware will be delivered to the Government under the award, there is potential unique item identification will be required IAW DFARS 252.211-7003 or similar verbiage.

xviii. Publication Approval/Disclosure of Information

Government review and approval will be required prior to dissemination or publication, except within and between the awardee and its subcontractors (if any), of classified and non- fundamental information developed under this award or contained in reports to be furnished pursuant to this award.

xix. Animal Welfare

Awards involving research, development, test, evaluation, or training on vertebrate animals will incorporate DFARS clause 252.235-7002 or similar verbiage.

xx. Protection of Human Subjects

Effective 29 July 2009, awards including or potentially including research involving human subjects IAW 32 CFR Part 219, DoD Directive 3216.02, and 10 U.S.C. 980, including research meeting exemption criteria under 32 CFR 219.101(b), will incorporate DFARS clause 252.235-7004 or similar verbiage.

All research involving human subjects, to include use of human biological specimens and human data, shall comply with the applicable federal and state laws and DoD policy/guidelines for human subject protection.

Institutions to be awarded funding for research involving human subjects (RIHS) must provide documentation of a current Federal Assurance of Compliance with Federal regulations for human subject protection, for example a Department of Health and Human Services, Office for Human Research Protections Federalwide Assurance (http://www.hhs.gov/ohrp). Additional Federal Assurance documentation may also be requested by the DOD. All institutions engaged in human subject research, to include subcontractor, must also have a valid Assurance. In additional, personnel involved in human subject research must provide documentation of completing appropriate training for the protection of human subjects.

As a condition of award, institutions must also provide documentation of Institutional Review Board (IRB) approval or a determination from an appropriate official in the institution that the work meets one of the exemption criteria in 32 C.F.R. 219. As part of the IRB review process, evidence of appropriate training for all investigators will need to be provided.

xxi. E-Verify

Contracts exceeding the simplified acquisition threshold may include the FAR clause 52.222- 54 "Employment Eligibility Verification" unless exempted by the conditions listed at FAR 22.1803.

i. Similar verbiage will be included in OT awards unless determined exempt through comparable determination.

xxii. Cybersecurity

Awardees are required to provide adequate security on all covered business information systems. Specific security requirements are listed in DFARS 252.204.7012, and compliance is mandatory. Note, DFARS clauses 252.204-7019 and 252.204-7020 also apply to this solicitation.

xxiii. Safeguarding Covered Defense Information Control

As prescribed in DFARS 252.204- 7008, for covered contractor information systems that are not part of an information technology service or system operated on behalf of the Government, the SBC represents that it will implement the security requirements specified by National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171, "Protecting Controlled Unclassified Information in Nonfederal Information Systems and Organizations".

xxiv. Limitations on the Use or Disclosure of Third- Party Contractor Reported Cyber Incident Information

As required in DFARS 252.204-7009, the Contractor must agree that certain conditions apply to any information it receives or creates in the performance of a resulting contract that is information obtained from a third-party's reporting of a cyber incident pursuant to DFARS clause 252.204-7012, Safeguarding Covered Defense Information and Cyber Incident Reporting (or derived from such information obtained under that clause).

xxv. Basic Safeguarding of Covered Contractor Information Systems

FAR 52.204-21, Basic Safeguarding of Covered Contractor Information Systems, is incorporated into this solicitation. In accordance with FAR 52.204-21, the contractor shall apply basic safeguarding requirements and procedures when the contractor or a subcontractor at any tier may have Federal contract information residing in or transiting through its information system.

xxvi. Prohibition on Contracting with Persons Who Have Business Operations with the Maduro Regime

DFARS 252.225-7055, Representation Regarding Business Operations with the Maduro Regime, is incorporated into this solicitation. In accordance with section 890 of the National Defense Authorization Act for Fiscal Year 2020 (Pub. L. 116-92), DoD is prohibited from entering into a contract for the procurement of products or services with any person that has business operations with an authority of the government of Venezuela that is not recognized as the legitimate government of Venezuela by the United States Government, unless the person has a valid license to operate in Venezuela issued by the Office of Foreign Assets Control of the Department of the Treasury.

xxvii. Prohibition on a ByteDance Covered Application

FAR 52.204-27, Prohibition on a ByteDance Covered Application, is incorporated into this solicitation. In accordance with FAR 52.204-27, contractors are prohibited from having or using a covered application on any information technology owned or managed by the Government, or on any information technology used or provided by contractors under contracts, including on equipment provided by the Contractor's employees.

e. Agency Recovery and Ongoing Reporting

In accordance with Section 5 of the SBIR and STTR Extension Act of 2022, the DOD will-

1) Require a small business concern receiving an award under its SBIR program to repay all amounts received from the DOD under the award if—
   - the small business concern makes a material misstatement that the DOD determines poses a risk to national security; or

- there is a change in ownership, change to entity structure, or other substantial change in circumstances of the small business concern that the Federal DOD determines poses a risk to national security; and

2) Require a small business concern receiving an award under its SBIR program to regularly report to the DOD and the Administration throughout the duration of the award on
    - any change to a disclosure required under the subparagraphs above;
    - any material misstatement made under the subparagraphs above; and
    - any change described above

f. Copyrights

With prior written permission of the contracting officer, the awardee may copyright (consistent with any appropriate national security considerations) material developed with DoD support. DoD receives a royalty-free license for the Federal Government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

g. Patents

Awardees normally may retain the principal worldwide patent rights to any invention developed with U.S. Government support. The government receives a royalty-free license for its use, reserves the right to require the patent holder to license others in certain limited circumstances, and requires that anyone exclusively licensed to sell the invention in the United States must normally manufacture it domestically. 35 U.S.C. § 205 authorizes that the government will not make public any information disclosing a government-supported invention for a period of five years to allow the awardee to pursue a patent. See also Section 6.8, Invention Reporting.

h. Invention Reporting

SBIR awardees must report inventions to the DOD within two months of the inventor's report to the awardee, via either paper documentation submission, including fax, or through the Edison Invention Reporting System at www.iedison.gov.

i. Technical Data Rights

Rights in technical data, including software, developed under the terms of any award resulting from proposals submitted in response to this CSO generally remain with the contractor, except that the U.S. Government obtains a royalty-free license to use such technical data only for government purposes during the period commencing with contract award and ending not less than twenty years after that date. This data should be marked with the restrictive legend specified in DFARS 252.227-7018.

Upon expiration of the twenty-year restrictive license, the government has government purpose rights in the SBIR data. During the license period, the U.S. Government may not release or disclose SBIR data to any person other than its support services contractors except: (1) for evaluation purposes; (2) as expressly permitted by the contractor; or (3) a use, release, or disclosure that is necessary for emergency repair or overhaul government-operated items. See DFARS clause 252.227-7018.

j. Multiple Procurements Planned

The CSO is seeking varying technical/scientific approaches and/or varying and new technologies that would be responsive to the problem statement(s) and area(s) of interest in the topic. Multiple procurements are planned and anticipated to be awarded as a result of the topic. Each proposal is considered a separate procurement and will be evaluated on its own merit. The DoD may award all of the proposals, some of the proposals, or none of the proposals submitted. Any per-award or per-topic funding caps are budgetary estimates only, and more or less funding may become available. Funding decisions are made with complete disregard to the other awards under the same topic.

k. Treatment of Proposals Received

The DOD reserves the right to award all of the proposals, some of the proposals, or none of the proposals submitted. The Federal Government is not responsible for any monies expended by the applicant before award. The SBIR/STTR Programs are not substitutes for existing unsolicited proposal mechanisms. Unsolicited proposals are not accepted under the DoD SBIR/STTR Programs for Phases I or II.

l. CSO, Selection, and Award Protests

Interested parties have the right to protest via procedures in FAR Subpart 33.1; protests exclusively related to this CSO's terms must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil.

For protests filed with the Government Accountability Office (GAO), a copy of the protest shall be submitted to the email address listed above (pre-award ONLY) or DoD Service/Component POC (postselection/award decision ONLY) within one day of filing with the GAO. Protests of small business status of a selected proposing SBC may also be made to the SBA via the procedures in FAR § 19.302.

For the purposes of a protest related to a particular topic selection, non-selection or award decision, protests should be served to the point-of-contact (POC) listed in the instructions of the DoD Component that authored the topic.

m. Technical Questions Related to Prospective Applicant Proposed Technical Solution

Applicants may submit written questions through DSIP Topic Q&A at https://www.dodsbirsttr.mil/submissions/login. In Topic Q&A, all questions and answers are posted electronically for general viewing. Identifying information for the questioner and respondent is not posted.

Questions submitted through the Topic Q&A are limited to technical information related to improving the understanding of a topic's requirements. **Any other questions, such as those asking for advice or guidance on solution approach, or administrative questions, such as SBIR or STTR program eligibility, technical proposal/cost proposal structure and page count, budget and duration limitations, or proposal due date WILL NOT receive a response.** As indicated above, applicants have ample opportunities to seek clarification regarding solicitation terms and provisions prior to the solicitation closing.

VI. APPENDIX A: DEFINITIONS

Note, additional definitions applicable to the SBIR/STTR program are available in the SBIR/STTR Policy Directive located at https://www.sbir.gov/sites/default/files/SBA%20SBIR_STTR_POLICY_DIRECTIVE_May2023.pdf.

Commercialization

The process of developing products, processes, technologies, or services and the production and delivery (whether by the originating party or others) of the products, processes, technologies, or services for sale to or use by the Federal government or commercial markets.

Essentially Equivalent Work

Work that is substantially the same research, which is proposed for funding in more than one contract proposal or grant application submitted to the same Federal agency or submitted to two or more different Federal agencies for review and funding consideration; or work where a specific research objective and the research design for accomplishing the objective are the same or closely related to another proposal or award, regardless of the funding source.

Export Control
The International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, and the Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, will apply to all projects with military or dual-use applications that develop beyond fundamental research, which is basic and applied research ordinarily published and shared broadly within the scientific community. More information is available at https://www.pmddtc.state.gov/ddtc_public.

NOTE: Export control compliance statements found in this document are not meant to be all inclusive. **They do not remove any liability from the applicant to comply with applicable ITAR or EAR export control restrictions** or from informing the Government of any potential export restrictions as fundamental research and development efforts proceed.

Foreign Persons
A foreign person is any natural person who is not a “lawful permanent resident” as defined by 8 U.S.C. 1101(a)(20) or who is not a "protected individual” as defined by 8 U.S.C. 1324b(a)(3). It also means any foreign corporation, business association, partnership, trust, society, or any other entity or group that is not incorporated or organized to do business in the United States, as well as international organizations, foreign governments, and any agency or subdivision of foreign governments (e.g., diplomatic missions).

“Lawfully admitted for permanent residence” (or someone who is a “lawful permanent resident”) means the status of having been lawfully accorded the privilege of residing permanently in the United States as an immigrant in accordance with the immigration laws, such status not having changed.

"Protected individual’’ means an individual who (A) is a citizen or national of the United States, or (B) is an alien who is lawfully admitted for permanent residence, is granted the status of an alien lawfully admitted for temporary residence under 8 U.S.C. § 1160(a) or 8 U.S.C. § 1255a(a)(1), is admitted as a refugee under 8 U.S.C. § 1157, or is granted asylum under Section 8 U.S.C. § 1158; but does not include

(i) an alien who fails to apply for naturalization within six months of the date the alien first becomes eligible (by virtue of period of lawful permanent residence) to apply for naturalization or, if later, within six months after November 6, 1986, and
(ii) an alien who has applied on a timely basis, but has not been naturalized as a citizen within 2 years after the date of the application, unless the alien can establish that the alien is actively pursuing naturalization, except that time consumed in the Service's processing the application shall not be counted toward the 2-year period.

Funding Agreement
Any contract, grant, or cooperative agreement entered into between any Federal Agency and any small business concern for the performance of experimental, developmental, or research work, including products or services, funded in whole or in part by the Federal Government. Only contracts and other transaction authority (OTA) agreements will be used by the DOD for SBIR awards. Additionally, all awards made under this CSO will be Firm-Fixed-Price.

Research or Research and Development (R/R&D)
Any activity that is:
a. A systematic, intensive study directed toward greater knowledge or understanding of the subject studied.
b. A systematic study directed specifically toward applying new knowledge to meet a recognized need; or

c. A systematic application of knowledge toward the production of useful materials, devices, and systems or methods, including design, development, and improvement of prototypes and new processes to meet specific requirements.

Small Business Concern (SBC)
A concern that meets the requirements set forth in 13 C.F.R. § 121.702 (available here: https://www.ecfr.gov/current/title-13/chapter-I/part-121/subpart-A/subject-group-ECFRb7921b3fcf04228/section-121.702).
An SBC must satisfy the following conditions on the date of award:

a. Is organized for profit, with a place of business located in the United States, which operates primarily within the United States or which makes a significant contribution to the United States economy through payment of taxes or use of American products, materials or labor;
b. Is in the legal form of an individual proprietorship, partnership, limited liability company, corporation, joint venture, association, trust or cooperative, except that if the concern is a joint venture, each entity to the venture must meet the requirements set forth in paragraph (c) below;
c. Is more than 50% directly owned and controlled by one or more individuals (who are citizens or permanent resident aliens of the United States), other small business concerns (each of which is more than 50% directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States), or any combination of these; and
d. Has, including its affiliates, not more than 500 employees. (For explanation of affiliate, see www.sba.gov/size.)

DEPARTMENT OF THE AIR FORCE (DAF)
SMALL BUSINESS INNOVATION RESEARCH (SBIR)
COMMERCIAL SOLUTIONS OPENING (CSO)

AFX255-DPCSO6 RELEASE 9 MANUFACTURING CHALLENGE DIRECT-TO-PHASE-II (D2P2) FOCUSED OPEN TOPIC
TOPIC RELEASE INSTRUCTIONS (TRI)

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I. DoD X25.5 COMMERCIAL SOLUTIONS OPENING (CSO)

See the DoD X25.5 Commercial Solutions Opening (CSO) for SBIR information specific to the Open Topic program and applicable laws, policies, and regulations.

The DAF intends these proposal submission instructions to clarify the Department of Defense (DoD) X25.5 Commercial Solutions Opening (CSO) as it applies to the topics solicited herein. **Firms must ensure proposals meet all requirements of the SBIR Program CSO posted on the DoD SBIR/STTR Innovation Portal (DSIP) at the proposal submission deadline date/time.**

II. DAF X25.5 RELEASE 9 DIRECT-TO-PHASE-II (D2P2) TOPIC RELEASE INSTRUCTIONS (TRI)

Release Number	Topic Number	Topic Title	Anticipated Number of Awards	Release Dates, Funding, Period of Performance	Agency POCs
X25.5 Release 9	AFX255-DPCSO6	Focused Open Topic Manufacturing Challenge Direct-to-Phase-II with a Clear Air/Space Force Stakeholder need	15	**Open:** June 4, 2025 **Close:** July 9, 2025 at 12:00pm Noon ET Maximum SBIR Funding Amount: $1,750,000 Maximum Period of Performance: 15 months	**(Primary)** AFWERX Open Topic Contracting Officer Ryan DeBonis (ryan.debonis.1@us.af.mil) and solicitations@afwerx.af.mil (Alternate) Air Force SBIR/STTR Contracting Officer Daniel Brewer (daniel.brewer.13@us.af.mil)

a. GENERAL TOPIC INFORMATION

i. Solicitation Topics Information

There is one (1) topic covered within this TRI. Manufacturing Challenge AFX255-DPCSO6 is specific to Air Force and Space Force needs.

A Small Business Concern (SBC) may only submit one (1) proposal under this topic release. If more than one proposal from an applicant is received, only the most recent proposal to be certified and submitted prior to the submission deadline will receive an evaluation.

ii. Due Date

The DAF will begin accepting proposals on June 04, 2025. **All proposals must be submitted by July 9, 2025, at 12:00 p.m. (noon) EST**. No exceptions will be made to the above submission deadlines. Proposals submitted after this date and time will not be considered for award.

iii. Number of Awards

The DAF currently anticipates making **15** awards under this topic release. Any per-award or per-topic funding caps are budgetary estimates only, and more or less funding may become available. The TRI is seeking varying technical/scientific approaches and/or varying and new technologies that would be responsive to the problem statement(s) and area(s) of interest in the topic. Multiple procurements are planned and anticipated to be awarded as a result of the topic, each proposal is considered a separate procurement and will be evaluated on its own merit, and that the Government may award all, some, or none of the proposals. Funding decisions are made with complete disregard to the other awards under the same topic.

iv. Maximum SBIR Funding Amount

Proposals that exceed $1,750,000.00 in value will not be considered for award. This amount does not include validated matching non-SBIR/STTR funding, either through private matching funds or non-SBIR/STTR governmental funding secured from a non-AFWERX entity.

v. Maximum Period of Performance

Proposals that exceed 15 months in duration will not be considered for award. Note: this is the TOTAL duration of the award. The DAF will not make distinctions between "technical" and "reporting" periods.

b. QUESTIONS ABOUT THIS TOPIC RELEASE AND TOPIC RELEASE INSTRUCTIONS

i. Questions and Answers Open Periods

There are dedicated periods for submitting questions related to the solicitation (CSO/TRI) and questions related to DSIP application submission.

The subject line of all submitted questions must identify the TRI to which you are inquiring.

ii. Solicitation Terms and Conditions Questions

Focus: Applicants may request clarification on the terms, conditions, and/or requirements of the solicitation, including the Government Engagement Memorandum and all solicitation attachments.
Open Period: 04 June 2025 - 18 June 2025, 4:00pm Eastern Time
Submission Instructions: Questions shall be emailed to ryan.debonis.1@us.af.mil, daniel.brewer.13@us.af.mil, and solicitations@afwerx.af.mil.
Government Responses: DAF intends to release a consolidated list of questions received and their respective answers on approximately 20 June 2025. DAF reserves the right not to answer every question submitted and will not answer any questions asking for feedback on the content of a specific applicant's proposal. All submitted questions will be kept anonymous.

DAF does not intend to answer questions received late.

iii. Administrative Questions

Focus: Administrative questions based on the process, not the interpretation of solicitation terms and conditions.
Open Period: Solicitation open period
Questions shall be emailed to: ryan.debonis.1@us.af.mil, daniel.brewer.13@us.af.mil, and solicitations@afwerx.af.mil.

Responses: Applicants will receive direct email responses. The Air Force will not answer questions that pertain solely to the DSIP application.

iv. DSIP Related Questions

Focus: Issues related to the DSIP website navigation and proposal submission processes that pertain directly to the DSIP system.
Open Period: Solicitation open period
Email questions to: DoDSBIRSupport@reisystems.com
Responses: DSIP helpdesk will respond directly to applicant questions.

v. Webform Related Questions

Focus: Issues related to accessing and submitting the webform and questions that pertain directly to the webform.
Open Period: June 4, 2025 - June 18, 2025 at 12:00pm Noon ET
Email questions to: rg.mfg.ext@afwerx.af.mil
Responses: Applicants will receive email responses.

c. SELECTION METHODOLOGY AND EVALUATION CRITERIA

All Phase II proposals will be evaluated on a competitive basis. Proposals will be peer reviewed based on the criteria below. Each proposal will be evaluated on its own merits. Evaluators will base their conclusions only on information contained in the proposal. It cannot be assumed that evaluators are acquainted with the small business concern or key individuals or any referenced experiments. Selections will represent the best value to the Government.

The following three (3) non-cost/price factors will be considered in descending order of importance:

i. Defense Need

The Defense Need being fulfilled and ability of the proposal to impact that Defense Need

a. The appropriateness, relevance, and specificity of an identified Defense Need. A complete and compelling Government Engagement Memorandum using the mandatory attachment, signed by an appropriate customer indicates the appropriateness, relevance, and specificity of an identified Defense Need. While not required, signed letters of support from other government personnel may be indicators of the appropriateness, relevance, and specificity of an identified Defense Need.
b. Adequacy of the proposed effort and its relationship to fulfilling the identified Defense Need.
c. The level of defense mission impact and urgency of the Defense Need.

ii. Technical Approach

a. The clarity of the problem statement, technical approach, soundness, technical merit, and innovation of the proposed technical approach.
b. The differentiation from current customer alternatives.
c. Includes the proposed Principal Investigators'/Project Managers', supporting staff, and consultants' qualifications to execute the proposed approach.

iii. Commercialization

The potential for government or private sector commercialization and benefits expected to accrue from commercialization.

a. The Small Business Concern's (SBC) record of commercializing SBIR or other research, as indicated by pre-sales, pilots, sales, revenue, active users, subscriptions, downloads, and/or other forms of traction/adoption.
b. The size of the market and potential for revenue.
c. The clarity of the SBC's business plan.
e. The existence of Phase II follow-on commitments for the subject research, and funding commitments related to the subject effort, whether from government sources, as documented in the DAF Government Engagement Memorandum (GEM) and/or the Volume 5 Phase II Funding Commitment Form, or private sources, as documented through a letter of commitment, are evidence of commercialization potential.
f. A sound transition strategy in Volume 2, *Commercialization Plan Outline*, is also evidence of commercialization potential.

All evaluation criteria, other than cost or price, when combined, are significantly more important than cost or price. Where technical evaluations are essentially equal in merit, cost and/or price will be considered in determining successful applicants.

The DAF will accept proposals from Small Business Concerns (SBCs) that are owned in majority by multiple venture capital operating companies, hedge funds, or private equity firms in response to this solicitation. In accordance with 15 United States Code (U.S.C). 638(dd), the DAF is prohibited from awarding more than 15% of its SBIR budget to SBCs with the previously mentioned ownership structures. In addition to the criteria above, once the value of selected proposals from such SBCs reaches 15% of the DAF's SBIR budget, proposals from such SBCs will no longer be considered for award.

Price reasonableness and realism will be considered to the extent appropriate. IAW 15 USC 638(dd)(7), investment by venture capital, private equity firms, or hedge funds shall not be used as an award criterion.

Other Factors Considered During the Selection Process:

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of Defense, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by small business concerns seeking a Federally funded award. The full text of the SBIR and STTR Extension Act of 2022 is available at: https://www.congress.gov/117/plaws/publ183/PLAW-117publ183.pdf.

In accordance with 15 USC 638(vv) (Section 4 of the SBIR and STTR Extension Act of 2022); the Deputy Secretary of Defense Memorandum; Subject: Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program dated May 13th, 2024; and the Department of the Air Force Guidance Memorandum to AFI 61-102, on DAF Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Due Diligence Program dated February 25, 2025, the Department of Defense will review all proposals submitted in response to this CSO to assess security risks presented by small business concerns seeking a Federally funded award. The Department

will use information provided by the small business concern in in response to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries and the proposal to conduct a risk-based due diligence review on the cybersecurity practices, patent analysis, employee analysis, and foreign ownership of a small business concern, including the financial ties and obligations (which shall include surety, equity, and debt obligations) of the small business concern and employees of the small business concern to a foreign country, foreign person, or foreign entity. The Department will also assess proposals utilizing open-source analysis and analytical tools, for the nondisclosures of the information set forth in 15 U.S.C. 638(g)(13). If DoD assesses that a small business concern has security risk(s), DoD will review the proposal, the evaluation, and the security risks and may choose to either 1) create a plan to mitigate the risk(s) or 2) DoD may decide to not select the proposal for award based upon a totality of the review.

The DAF will also utilize a due diligence process as outlined in 15 U.S.C. 638(vv) to assess proposals utilizing open-source analysis and analytical tools. If DAF assesses that a small business concern has security risk(s), DoD will review the proposal, the evaluation, and the security risks and may choose not to select the proposal for award based upon a review of the totality of the circumstances.

Research or Research & Development: Consistent with 15 USC 638(e)(4) all awards under this solicitation must constitute "research" or "research and development" as defined in 15 USC 638(e)(5). Proposals determined not to constitute "research", or "research and development" will be disqualified and no longer considered for award.

d. COVER SHEET (VOLUME 1)

The Cover Sheet (Volume 1) is prepared on DSIP and in accordance with the DoD Program CSO.

e. TECHNICAL VOLUME (VOLUME 2)

Table 1: Technical Volume Content Description (Volume 2)	
Item	**Description**
Type of File	The Technical Volume must be a single Portable Document Format (.pdf) file, including graphics. Perform a virus check before uploading the Technical Volume file. If a virus is detected, it may cause rejection of the proposal. Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document. Do not include hyperlinks or any form of active link. Information linked to the proposal in such a manner will not be considered as part of the Government's evaluation. **Please note that files submitted as part of Volume 2 have a file size limit of 15 MB.**
Length Limitation	The White Paper is limited to 15 pages; any pages in excess of 15 will not be considered. The Coversheet, Glossary, and Table of Contents do NOT count against the 15-page limit.

Layout	Number all pages consecutively. Each page should contain the company name, topic number, and proposal number assigned by DSIP when the Coversheet was created. Use no smaller than 10-point font.
Proprietary Information in Proposals	The following legend must appear on each page of the proposal that contains information the applicant wishes to protect: *Use or disclosure of information contained on this sheet is subject to the restriction on the title page of this proposal.*
Content	
Table of Contents	Include a Table of Contents immediately following the Proposal Coversheet. The Table of Contents does not count against the 15-page limit. Proposing SBCs that include data they do not want disclosed to the public for any purpose, or used by the Government except for evaluation purposes, shall mark the first page of the proposal submission with the following legend: *This proposal contains information that shall not be disclosed outside the federal government and shall not be duplicated, used, or disclosed in whole or in part for any purpose other than evaluation of this proposal, unless authorized by law. The Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract if award is made as a result of the submission of this proposal. The information subject to these restrictions are contained on all pages of the proposal except for pages [insert page numbers or other identification of pages that contain no restricted information].*
Glossary	Include a glossary of acronyms and abbreviations used in the proposal. The Glossary does not count against the 15-page limit.
Technical Summary	
Identification and Significance of the Problem or Opportunity	Briefly describe the specific problem/opportunity addressed in the feasibility study and its relationship to the Phase II effort.
Manufacturing Technology Focus Area Justification	To help evaluators understand how your proposed solution fits the Manufacturing Technology Focus Areas, provide a written justification that includes these three key parts: First, clearly identify which of the following six Manufacturing Technology Focus Areas your proposed solution directly addresses: Engine Production with Advanced Manufacturing, Advanced Composite Structures, Semiconductor Production Innovations, Radiation-Tolerant/Hardened Electronics, System Integration and Interoperability, and/or Affordable Energetics. Second, for each focus area you identified, thoroughly explain how your proposed technology and solution meet the specific definitions provided

	for that area. Your explanation should clearly show the connection between your innovation and the key aspects described in the definition. For instance, if you chose "Engine Production with Advanced Manufacturing," detail how your approach uses innovative manufacturing processes, materials, or automation to improve engine performance, efficiency, cost, or reliability for Air Force and Space Force applications. Third, ensure your justification offers enough detail and evidence to reasonably demonstrate that your proposed solution aligns with at least one of the defined focus areas. For this Manufacturing Technologies focused open topic, your application *must* clearly validate its fit within one or more of these six areas. **Applications that fail to reasonably validate your applications alignment with one of the six technology focus areas will be disqualified.**
Non-Defense Commercial Solution	Describe the non-Defense commercial solution, including its use by non-Defense commercial customers. Describe its technical specifications as well as the qualitative and quantitative value it brings to the solution's users. Relevant supporting data, such as journal articles, literature, government publications, etc., should be contained or referenced in the proposal and will count toward the page limit. Additionally, answer the following questions: ● Does the proposed solution represent an entirely novel standalone solution, or does it modify/build upon an existing product or service? ● Has the proposed solution been (i) sold, leased, or licensed to the public; OR (ii) offered for sale, lease, or license to the public? Provide quantifiable data evidencing this sale, lease, or licensing. ● What is the non-defense market opportunity of your proposed solution? Can this be quantified or described? ● Is the proposed item "of a type", i.e., similar to a commercial item, customarily used by the public or by non-governmental entities for non-governmental purposes? ● What are the end-user use cases for the proposed solution and how does the proposed solution fulfill these use cases? ● How is the proposed solution different from similar competitor solutions? ● What are the proposed solution's technical risks and how are they mitigated? ● What is the proposed solution's Technology Readiness Level (TRL), and have any previous results shown the technology's viability? Include, where applicable, pre-sales, pilots, sales, revenue, active users, subscriptions, downloads, and/or other

	forms of traction/adoption for the proposed solution. Additional information on TRLs can be found in the OUSD(R&E) Technology Readiness Assessment Guidebook at https://www.cto.mil/wp-content/uploads/2023/07/TRA-Guide-Jun2023.pdf. ● Broad adoption and commercial success can be indicators of commercial potential and reduced technical risk but are not mandated. **Clearly delineate between the proposed solution's commercial success and the overall company's success.** This is an RDT&E solicitation, not a Commercial-Off-the-Shelf (COTS) resale solicitation. See FAR 2.101 for a full definition of commercial items.
Proposed Adaptation of Non-Defense Commercial Solution	Describe the proposed Research/Research and Development (R/R&D) adaptation of, or the Test and Evaluation (T&E) (strictly for use in technical or weapons systems), to be performed on the non-Defense commercial solution to meet a US Government end-user need. In this section, answer the following questions: ● How similar is the modified item to others sold in the commercial marketplace to non-US government customers? ● Does the supplier perform similar modifications for non-US government customers? ● Do DAF-unique modifications change the product's essential use and purpose? ● Are there differences in the production/manufacturing/delivery processes used to perform the modification for the federal government versus non-government customers? ● What are the quantitative benefits expected for DAF end-users? ● When adapting the commercial solution, what is the intended COTS or commercial product/process modification? Explain. ● If applicable, what T&E will be performed on the solution, and what Technical or Weapons System will the solution be integrated into? What is the desired outcome of that T&E?
Phase I-Type "Feasibility Study" Results	The Proposal shall detail the proposed solution's ability to meet DAF end-users' and customers' needs. These results shall be framed in the context of DAF End-User and Customer Exploration performed for the effort prior to this effort. As stated in the DoD CSO X25.5 Preface specific to D2P2 proposals, **"Feasibility documentation cannot be based upon or logically extend from any prior or ongoing federally funded SBIR or STTR work. The applicant must provide adequate documentation clearly demonstrating the Phase I-type feasibility study is not based upon or logically extended from any prior or ongoing federally funded SBIR or STTR work.**

	● Scientific or Technical R/R&D effort, including research questions, methods, results, and relevant literature. ● DAF End-User and Customer Exploration Methods: Describe methods, both successful and unsuccessful, used to perform technical and scientific feasibility analysis, as well as DAF Customer and End-User Exploration, for adapting the non-Defense commercial solution to a specific DAF end-user's needs. This should include detailed reports of at least one (1) in-depth interview with a DAF end-user and one (1) in-depth interview with a DAF customer. ● Empower and Committed DAF End-Users: list and describe specific DAF end-users (e.g., operators) willing to support further R&D, testing, evaluation, or trial for the proposed Phase II solution. ● Specifically explain how the proposed solution meets the end-user's needs. Describe in as much detail as possible the firm's understanding of the DAF end-user's problem or benefit area. ● Empowered and Committed DAF Customers: list and describe the specific DAF customer (e.g., a Program Manager at a System Program Office) willing to assist with transitioning the proposed D2P2 solution to a Phase III or other non-SBIR contract vehicle. ● Joint organizations may be the end-user or customer, but the Government Engagement Memorandum must clearly articulate how the joint end-user/customer represents a validated DAF need. In these cases, the Primary TPOC must be from the DAF. ● Empowered and Committed US Non-DAF Government End-Users: commitments from US non-DAF government end-users will also be considered. However, there MUST be at least one (1) DAF/joint end-user and customer, as described above. ● Inclusion of a Government Engagement Memorandum is **mandatory** for all proposals. **Proposals without a signed Government Engagement Memorandum or proposals that do not follow the mandatory template will not be considered for award.** Government Engagement Memoranda must be uploaded to the "Letters of Support" Section in Volume 5, Supporting Documents.
Compliance and Regulatory Activities	Completion of Attachment 2, *Regulatory Compliance Initial Safety Assessment* form, in its entirety, is required as part of the proposal submission (included in Volume 5).
Phase II Technical Objectives and Key Results	Clearly describe three (3) to five (5) objectives of the Phase II RDT&E effort. These objectives should be tied to specific proposed Phase II trial tasks. These objectives shall be qualitative and specific to the participating DAF customers. The objectives shall be tied to specific time periods, tangible and unambiguous regarding anticipated results. These objectives shall describe end-state outcomes (i.e. what will be done), rather than processes or activities (i.e., how it will be done). Each objective shall be accompanied by three (3) to five (5) specific 'key

	results', measurable throughout Phase II performance. These key results will likely be quantitative in nature, e.g., deliver X widgets to DAF customer. Non-quantitative key results shall be clearly measurable. **Required Stakeholders Needed to Accomplish Phase II:** List anticipated interactions with stakeholders necessary to accomplish the Phase II trial. Such stakeholders may include lawyers, test range officials, information assurance officials, system program office engineers, etc. Clearly describe Phase II trial completion requirements not within the applicant's immediate control and plans to work within those constraints.
Phase II Work Plan Outline	Outline provided in Table 2.
Commercialization Strategy	Outline provided in Table 3.
Military Applications and DoD Customers	Indicate how your firm believes the technology development under this proposal could contribute to future mission-need fulfillment. Describe the mission-impact of your proposed effort in terms of impacts on the DAF and DoD. State how your technology development aligns with your identified customer. You may choose to work with your customer to refine the defense need of your proposal and provide impact statements. Identify the DoD agency/organization most likely to become the solution's customer. In many cases, the potential DoD customer will be different from the end user. For example, the user may be in an operational flying unit and the customer in a System Program Office for the related aircraft. State whether a DoD agency has expressed interest in, or commitment to, a non-SBIR, federally funded Phase III effort or a Phase II enhancement with matching SBIR funds. If so, please include relevant details regarding funding availability, e.g., fiscal year, for solution transition, and other relevant funding details, such as funding type, e.g., RDT&E, Operations and Maintenance, etc. Include agency points of contact names, email addresses, and telephone numbers.
Foreign Persons	Identify all foreign persons **and** non-U.S. citizens **and** dual citizens proposed as direct employees, subcontractors, or consultants. For these individuals, in addition to technical resumes, please provide countries of origin, type of visas or work permits held, and an explanation of their anticipated involvement level in the project, i.e., tasks to be assigned. When the proposed technology is subject to export control, these individuals, if permitted to participate, are limited to work in the public domain. Further, tasks assigned must not be capable of assimilation into an understanding of the project's overall objectives. This precludes foreign nationals from acting in key positions such as Principal Investigator, Senior Engineer, etc. Additional information may be requested during negotiations to verify a foreign person's eligibility. If the proposed work is International Traffic in Arms Regulations (ITAR) Restricted or otherwise export-controlled, the information and materials provided pursuant to or resulting from that proposal are

	restricted under the ITAR, 22 CFR Parts 120 - 130, which controls the export of defense-related material and services, including the export of sensitive technical data. Non-U.S. citizens may perform work under an award resulting from these topics only if they are "Lawfully Admitted for Permanent Residence", as defined in 8 U.S.C. 1101(20) or are designated as "Protected Individuals" as defined by 8 U.S.C. 1324b(a)(3). If a proposal for an ITAR-restricted technology involves participation of a non-U.S. citizen who is not in one of the above two categories, the proposal may be rejected.
Key Personnel	Identify key personnel involved in the project, including subcontractors and consultants. The applicant shall address qualifications for conducting RDT&E in addition to qualifications for commercialization. Include information regarding education, experience, and citizenship. A technical resume for the PI, including publications, if any, shall be included.
Non-Defense Commercial Customers	Past revenue from commercial customers of the proposed non-Defense commercial solution. Clearly list non-Defense commercial customers, including revenue received as a result of the proposed solution's sales. Do not include other solutions the company may offer. Product revenue is realized by directly selling a solution to solve a problem. Potential revenue-generating projects directly related to the proposed non-Defense commercial solution or adaptation thereof. Separately list potential revenue-generating projects with Defense or non-Defense customers directly related to the proposed solution(s). Note: On-going or completed projects/sales, such as those in section 2.5.1, are strongly preferred to potential projects.
Investors and Partners	Identify company investors, shareholders, or strategic advisors making significant contributions to support the Phase II effort. Explain how their past experiences or track record will support the technical or commercialization efforts. This could include leveraging other entities in company investors' portfolios or accessing other capital sources.
Related Work	Describe significant activities directly related to the proposed effort, including previous programs conducted by the Principal Investigator (PI), proposing SBC, consultants, or others, and their application to the proposed project. List reviewers providing comments regarding the applicant's knowledge of the state-of-the-art in the specific approach proposed. Specify related government efforts, SBIR or otherwise, awarded, previously proposed, or intended for proposal.

Table 2: Non-Proprietary Phase II Work Plan Outline (Included in Volume 2)	
Proposals including private investment cost match should include the tasks to be supported with those funds in the Work Plan. Tasks to be executed with private investment should be discernible from SBIR-funded tasks. If awarded, the SBIR-funded and private investment-funded tasks are included in different parts of the award document. The Work Plan should be structured around the effort's purpose and requirements set forth in clear, specific, and objective terms with measurable outcomes. The work plan has two (2) mandatory features; (1) Work stated in terms of outcomes or results, rather than performance methods, and (2) measurable performance standards, including assessment methods for these standards. The plan should include the location and mechanics for accomplishing the proposed approach. Phase II projects shall attempt to demonstrate the Phase I-like effort's technical feasibility, including transition results to the private sector or government customer.	
Non-Proprietary Work Plan Instructions	
Scope	List the effort's major requirements and specifications.
Task Outline	Provide an outline of work to be accomplished throughout the effort. For proposals with government funding commitments (where the funds will be sent to AFRL/RGF to be included in the Phase II effort), clearly identify the tasks that will be funded by the SBIR program and the tasks that will be funded by the government organization that is committing funds. For those proposals that implicate **Research Involving Human Subjects (RIHS)**, specifically indicate which tasks fit the definition of RIHS. The amount of time required for the IRB to review and approve the protocol will vary depending on such factors as the IRB's procedures, the complexity of the research, the level of risk to study participants, and the responsiveness of the investigator. The average IRB approval process can last between one (1) and three (3) months. Once the IRB has approved the research, the DAF will review the protocol and the IRB's determination to ensure that the research will be conducted in compliance with DAF policies. The DAF review process can last between three (3) to six (6) months. Ample time should be allotted in the schedule to complete both the IRB and DoD approval processes prior to recruiting subjects. **No funding can be used towards human subject research until ALL approvals are granted. Applicants proposing research involving human and/or animal use are encouraged to separate those tasks in the technical proposal and cost proposal in order to avoid potential delay of contract award.**
Deliverables	Include a section clearly describing the specific sample/prototype hardware/software to be delivered, as well as data deliverables, schedules, and quantities.
Unique Item Identification	DFARS 252.211-7003, Item Identification and Valuation, may be required. Applicants should provide a unique item identifier (UID) for all items for which the Government's unit acquisition cost is $5,000.00 or more.
	Required Reports

At a minimum, the following reports are required for ALL Phase II awards and must be included in the Work Plan. All are considered scientific and technical reports.	
Final Report	A completed SF 298, *Report Documentation Page*, will be used as the report's first page. This will include identifying the purpose of the work, providing a brief description of the effort accomplished, and listing potential results applications. The summary may be published by DoD. Therefore, it shall not contain proprietary information. It shall also contain project objectives met, work completed, results obtained, and technical feasibility estimates. This report shall be no more than 25 pages.
Status Reports	Status reports are due quarterly, at a minimum. These shall be concise documents describing progress in meeting the selected DAF end-user's needs. Each status report shall be no longer than 15 slides. The status reports shall include progress toward Phase II objectives and key results, as well as a quantitative measure between zero (0) and one (1) of progress toward each.
Phase II Summary Report	The Phase II summary report is due at technical effort completion. It shall be submitted via email to the cognizant AF Contracting Officer/Agreement Officer (CO/AO). Reports shall not exceed 700 words, including a description of the technology and anticipated applications/benefits for government and/or private sector use.
Safety-Related Deliverables	Safety requirements shall be followed in the proposed project's design and performance. The Work Plan must contain separate sections specifically addressing the following: ● Preliminary Hazard Analysis Report (if applicable): If hardware is to be developed, the contract shall include at least a preliminary hazard analysis included in the deliverables. ● Hazardous Material Report: If use of hazardous material is anticipated, the following are required: materials identification; materials purpose; and possible alternative/procedures/safeguards to minimize risk.
Additional Reporting	The DAF end-user or customer may require additional reporting or documentation including: ● Software documentation and user manuals ● Engineering drawings ● Operation and Maintenance documentation ● Safety hazard analysis when the project will result in partial or total development/delivery of hardware ● Updated commercialization results
Funding Commitments Guidance	The final Phase II status report must include a brief accounting, in company format, regarding investor funds' expenditures to support the project.

Milestone Schedule	**Include a completed milestone schedule.** Each milestone marks the completion of a required deliverable (e.g., prototype component, test plan, prototype production, final report submission, etc.). Status reports cannot be milestones but are deliverables as described below. Describe how the milestones will be identified as completed. Milestone payments shall reflect justifiable costs to achieve completion. Supporting rationale (e.g., invoices from non-Defense commercial customers), will be included in Volume 5, Supporting Documents. Note: Pricing information from non-Defense commercial customers is likely proprietary and should be marked as such. **Instructions:** The milestone schedule must indicate the following fields: Contract Line Item Number (CLIN), delivery date, CLIN description (listed in terms of separate tasks, deliverables, and acceptance criteria), and payment amount (listed in whole dollars). **Data Field Instructions:** **CLIN:** Entered as a four-digit entry, such as 0009 or 0010. **There should be a minimum of eight (8) and a maximum of twelve (12) CLINs that correspond with the applicant's proposed milestones.** Deviations for proposals with less than eight (8) or more than twelve (12) CLINs are negotiable if selected for award. **Delivery Date** (listed as months after contract award)**:** Entries should be represented as "Award + X months". **Ensure the final delivery date does not exceed the maximum period of performance identified in these Topic Release Instructions.** **CLIN Description** (must include the following elements)**:** **- Task:** This shall be the title of the CLIN and should represent the types of work being conducted during this portion of the effort. **- Deliverables**: Enter the specific deliverables that will be delivered by the delivery date previously listed for this CLIN. You may list as many deliverables as necessary to accurately convey what will be provided to the Government for acceptance during this portion of the effort. **You may not list quarterly status reports (defined below) as a standalone CLIN. Status reports may be listed as a deliverable but must be accompanied by another deliverable eligible for separate payment under the CLIN.** **- Acceptance Criteria:** Acceptable verbiage for all CLINs may be, "The TPOC confirms that the delivered data, reports, hardware,

	prototypes and/or presentations are accurate, complete and in accordance with the contract". Applicants may add to or modify the recommended verbiage, if necessary, to better align with the proposed deliverable. **Payment:** Insert the whole dollar amount proposed for the work completed/delivered under the applicable CLIN. Non-SBIR federal funds must be identified. Do not include Phase III/private funds. **Note: Consistent with the below language, all proposals shall include the delivery of a kick-off meeting presentation for CLIN 0001. Deviations from the below verbiage are negotiable if selected for award.** **CLIN 0001** **Delivery Date:** "Award + 1 month" **CLIN Deliverable:** "Submission of kick-off meeting presentation using the mandatory template provided through the Virtual Contract Management Site (VCMS) in PDF format." VCMS access will be explained in the Welcome Letter accompanying contract award. **Acceptance Criteria:** TPOC confirmation that the delivered kick-off presentation is complete, accurate, and in accordance with the contract. **Payment:** The payment amount for the first CLIN should not exceed 5% of the total contract value. Any deviations from this amount are negotiable if selected for award. **Any deviations identified above, do not need to be explained in your proposal submission. Applicants should be prepared to discuss deviations with the Contracting Officer during negotiations if the applicant's proposal is selected for negotiations.** **See example Milestone Schedule chart below.**

MILESTONE CHART (EXAMPLE)

CLIN/MILESTONE NUMBER Minimum 8 and maximum 12	**EXPECTED DELIVERY** (month(s) after contract award) Ensure you do not exceed the Topic Release Instructions stated maximum period	**CLIN/MILESTONE DELIVERABLE**	**PAYMENT** Non-SBIR Federal funds must be identified. Do not include Phase III/Private funds.

	of performance		
(EXAMPLE) 0001 (Required)	Award + 1 month (Required)	Task (Required): Project Kick-off Meeting and Presentation Deliverable(s) (Required): Submission of kickoff meeting presentation using the mandatory template provided through the Ventures Contract Management Site (VCMS) in PDF format. Acceptance Criteria: The TPOC confirms that the delivered kick-off presentation is complete, accurate, and in accordance with the contract.	Enter your whole dollar amount here. (required: not to exceed 5% of total)
(EXAMPLE) 0002	Award + 3 months	Task: Produce, Test, and Demonstrate Iteration #1 Deliverable(s): The awardee shall deliver two working V1.0 prototypes of the Flux Capacitor along with the detailed design package drawings. Acceptance Criteria: The TPOC confirms that the delivered data, reports, hardware, prototypes and/ or presentations are accurate, complete and in accordance with the contract.	Enter your whole dollar amount here. Example: $99,759.00 (DAF Customer 3600)
(EXAMPLE) 0003	Award + 6 months	Task: Produce, Test, and Demonstrate Iteration #2 Deliverable(s) (Required): The	Enter your whole dollar amount here. Example: $251,226.00

		awardee shall deliver one enhanced V2.0 prototype that meets the specifications XYZ. The awardee shall deliver the Test Plan. Acceptance Criteria: The TPOC confirms that the delivered data, reports, hardware, prototypes and/or presentations are accurate, complete and in accordance with the contract.	
Required Final Milestone (CLIN 0008 - 0012)	Award + 15 months		Enter your whole dollar amount here.

Table 3: Commercialization Plan Outline (Included in Volume 2)	
The SBA and DAF require Phase II proposals include a commercialization plan. The plan should briefly describe the commercialization potential for the proposed project's anticipated results, as well as plans for exploitation in both the Defense and non-Defense markets. The plan must include the items below:	
Company Information	Focused objectives/core competencies; specialization area(s); products with significant sales; and history of previous federal and non-federal funding, regulatory experience, and subsequent commercialization.
Customer and Competition	Clear description of key technology objectives, current competition, and advantages compared to competing products or services; description of hurdles to innovation acceptance.
Market	Milestones, target dates, market size analysis, and estimated market share after first year and fifth year sales; plan to obtain market share.
Intellectual Property	Patent status, technology lead, trade secrets, or other demonstration of a plan to achieve sufficient protection to realize the commercialization stage and attain at least a temporal competitive advantage.
Financing	Plans for securing further necessary funding.
Assistance and Mentoring	Plans for securing needed technical or business assistance through mentoring, partnering, or through arrangements with State assistance programs, Small Business Development Centers, FFRDCs, Manufacturing Extension Partnership centers, the DAF Mentor-Protégé Program, or other assistance providers. Specifically address the following questions: ● What is the first product into which the technology will be incorporated? ● Who are the customers and what is the estimated market size? How much money is needed to bring the technology to market? How will the funding be raised? ● Does the company possess marketing expertise? If not, how will it be obtained? ● What companies are your competitors, and what is the SBC's price and/or quality advantage over them? ● Are there private sector or non-SBIR/STTR funding sources demonstrating commitment to Phase II results? This includes both outside private investment such as venture capital, angel investors, state or local government programs, etc., and internal funding sources, such as funding from affiliates. Please delineate past and potential future investments, e.g., currently raising a round or have signed term sheets. Also note if commitments are conditional on receiving a Phase II award.

	● Has your company received, or will it receive, any foreign investment? If so, what is the source and the received or anticipated amount? ● Are there Phase III (government or commercial) follow-on commitments for the proposed technology? ● Are there any other commercial potential indicators? Consider pre-sales, pilots, sales, revenue, active users, subscriptions, downloads, and/or other forms of traction/adoption and commercial signals of interest, demand, and faith in your team/solution. ● What is the last 12 months' total revenue from non-Defense commercial solution sales? State the proposed Phase II's anticipated end results, specifically, plans to transition to a Phase III with a potential government customer.

f. COST VOLUME (VOLUME 3)

Table 4: Cost Volume Description (Volume 3)	
Complete the Cost Volume as shown in the Cost Breakdown Guidance. This is accomplished by using the online cost volume form in DSIP. Some items may not apply to the proposed project. If not, do not provide information for those items. Enough information shall be provided for DAF to understand the company's plan to utilize the requested funds, i.e., the purpose, necessity, and reasonableness of each expenditure. Provide sufficient detail, as found below, regarding funds' use if the contract is awarded. The itemized listing may be placed in the "Explanatory Material" section of the on-line Cost Volume (if there is enough room) or submitted in Volume 5, Supporting Documents, under the "Other" dropdown option. Note: Only one (1) file can be uploaded to DSIP. Note: If no exceptions are taken to an applicant's proposal, the Government **may** make award without further communications. Therefore, the initial proposal should contain an applicant's best terms from a cost/ price and technical standpoint.	
Measurable Milestones	Fixed price payments shall be tied to measurable milestones, as agreed to with the Government. For Phase II efforts including government second phase funding commitments wherein RGK receives non-SBIR, governmental funding for a Phase II effort, the cost proposal should clearly identify SBIR-funded tasks and non-SBIR-funded tasks. Government second phase funding commitments awarded as Phase IIIs and private investment should not be included in the Cost Volume.
Direct Labor Costs	For direct costs, include substantiating information such as vendor quotes, previous orders, historical data, etc. Costs without substantiating justification should be detailed through the cost estimating process. All substantiating evidence should be included in the *Additional Cost Information* section of Volume 5, Supporting Documents. If selected for award, the Government may require further documentation to substantiate costs.

	Identify key personnel by name, if possible, or by labor category if specific names are not available. The number of hours, labor overhead and/or fringe benefits, and actual hourly rates for each individual are required.
Direct Material Costs	Provide an itemized list, including types, quantities, price, and, when appropriate, purpose, for materials, parts, and supplies. If proposing direct material, justification information, including vendor quotes, historical costs, etc., can be included as *Supplemental Cost Information* in Volume 5, Supporting Documents.
Other Direct Costs	
This category of costs includes specialized services such as machining or milling, special testing or analysis, and/or cost incurred in obtaining temporary use of specialized equipment. Proposals, which include leased hardware, must provide an adequate lease vs. purchase justification or rationale. If proposing other direct costs, additional information can be included as *Supplemental Cost Information* in Volume 5, Supporting Documents.	
Travel	Include travel costs in *Other Direct Costs*. Travel costs must relate to the needs of the project. Break out travel cost by trip, number of travelers, airfare, per diem, lodging, etc. The number of trips required, as well as the destination and purpose of each should be included. Recommend budgeting at least one (1) trip to the Air Force location managing the contract. Estimation of travel costs need not be an overly onerous exercise, and actual travel cost may differ from your proposed travel. This information allows the government Contracting Officer (CO) to ensure travel costs are not excessive.
Subcontracts	Include subcontractor/consultant costs in *Other Direct Costs*. Involvement of university or other consultants in the planning and/or research stages of the project may be appropriate. If so, describe in detail and include information in the Cost Volume. The proposed total of all consultant fees, facility leases or usage fees, and other subcontract or purchase agreements may not exceed one-half the total contract price or cost, unless otherwise approved in writing by the CO. Support subcontract costs with copies of the subcontract agreements. The supporting agreement documents must adequately describe the work to be performed. At a minimum, an applicant must include a Statement of Work (SOW) with a corresponding detailed Cost Volume for each planned subcontract.
Consultants	Provide separate letters of agreement for each consultant. The letter should briefly state what service or assistance will be provided, the number of hours required, and the hourly rate.
Special Tooling/Test Equipment/Material	The inclusion of equipment and materials will be carefully reviewed relative to need and appropriateness of the work proposed. The purchase of special tooling and test equipment must, in the opinion of

	the CO, be advantageous to the Government and relate directly to the specific effort. Such items may include such items as innovative instrumentation and/or automatic test equipment. Title to property furnished or funded by the Government will be vested with the AF, unless determined that title transfer to the contractor is more cost effective than equipment recovery by the Government.

g. COMPANY COMMERCIALIZATION REPORT (VOLUME 4)

Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoD Program CSO for full details on this requirement. Information contained in the CCR will not be considered by DAF during proposal evaluations.

h. SUPPORTING DOCUMENTS (VOLUME 5)

Table 5: Supporting Documents (Volume 5)
This volume is used to submit additional documentation supporting the Technical Volume (Volume 2) and the Cost Volume (Volume 3). **Please note that files submitted as part of Volume 5 have a file size limit of 5 MB. When combining .pdf documents, ensure digital signatures are not stripped.**
Documents Required at Proposal Submission
Attachment 1: Government Engagement Memorandum
Attachment 2: Regulatory Compliance Initial Safety Assessment Form
Attachment 3: SBIR/STTR Funding Agreement Certification
Attachment 4: SBIR/STTR Funding Agreement Lifecycle Certification
Required (if applicable) Documents
Attachment 5: Phase II Funding Commitment Form
Attachment 6: Identification and Assertion of use, Release, or Disclosure Restrictions
Attachment 7: Certification for SBIR Applicants that are Majority-Owned by Multiple Venture Capital Operating Companies, Hedge Funds, or Private Equity Firms
Recommended at Proposal Submission

NIST SP 800-171 Assessment Requirements

Applicants are required to implement NIST SP 800-171 in accordance with DFARS 252.204-7019 to receive a contract award. Applicants shall have a current basic assessment (see DFARS 252.204-7020, NIST SP 800-171 DoD Assessment Requirements) for each covered contractor information system that is relevant to the offer and/or contract.

DAF does not require the basic assessment to be completed nor submitted with its proposal under the applicable solicitation to which they are applying. The NIST assessment requirements are contained in the CSO to provide applicants with advanced notice that should their application be Selected that they SHALL meet these requirements herein prior to a contract being awarded. Failure to meet the requirements may lead to disqualification during contract negotiations.

DAF encourages all applicants with current SBIR/STTR funded contracts to follow the instructions below to obtain a Procurement Integrated Enterprise Environment (PIEE) account to meet the NIST requirement within the Supplier Performance Risk System (SPRS) module in PIEE.

The Basic, Medium, and High NIST SP 800-171 DoD Assessments are described in the NIST SP 800-171 DoD Assessment Methodology located at https://dodprocurementtoolbox.com/site-pages/cybersecurity-policy-regulations.

In accordance with DFARS 252.204-7020, the SBC shall provide access to its facilities, systems, and personnel necessary for the Government to conduct a Medium or High NIST SP 800-171 DoD Assessment, as described in NIST SP 800-171 DoD Assessment Methodology, linked above.

Instructions for Completing NIST SP 800-171 Basic Assessment

- To complete and obtain a basic self-assessment score, go to www.projectsprectrum.io.
- Login or register to begin the assessment
- Once logged in, scroll to the bottom and click on "NIST 800-171"
- Upon completion, save the document, as you will need your self-assessment score and date submitted
- Submit NIST SP 800-171 in one of two ways:
  - **Preferred:** All applicants should obtain a Procurement Integrated Enterprise Environment (PIEE) account and submit their firm's assessment directly to the Supplier Performance Risk System (SPRS) module in PIEE. Failure to ensure your company's score is published in PIEE's SPRS module by **September 17, 2025**, as well as failure to provide any additional requested information, as specified in DFARS 252.204-7019, may result in disqualification. DAF recommends early completion and submission of this information. Instructions on creating a PIEE/SPRS account can be found at https://www.sprs.csd.disa.mil/pdf/NISTSP800-171QuickEntryGuide.pdf.
  - **Alternate:** Email WEBPTSMH.fct@navy.mil and include the required information specified in DFARS 252.204-7019(d)(1). Confirm with the email address above that your company's NIST SP 800-171 self-assessment score has been posted to the SPRS module in the PIEE.

Documents Acceptable for Inclusion in Volume 5, Supporting Documents	
Supplemental Cost Information	This document is used to provide additional cost information to augment the Cost Volume (Volume 3) provided in the online form. The Supplemental Cost Information detail must be adequate to enable Department of the Air Force personnel to determine the

	purpose, necessity, and reasonableness of each cost element provided in the Cost Volume (Volume 3). For significant costs, e.g. large number of hours, very high hourly wages, substantial labor overhead/fringe rates, etc., provide justification. A helpful resource for determining average rates is https://www.bls.gov/oes/current/oes_nat.htm.
Direct Material Costs Supporting Documentation	See Table 4, *Cost Volume Description (Volume 3)*.
Travel Included in Other Direct Costs	Include the sources of estimated costs for flights, lodging, meals, and ground transportation. A useful resource for this section is https://www.gsa.gov/travel.
Special Tooling/Test Equipment Costs	See Table 4, *Cost Volume Description (Volume 3)*.
Subcontractor Agreements	If the applicant intends to utilize consultants, subcontractors, academia, etc., describe the relationship in detail, including information in the Cost Volume (Volume 3). The proposed total of all consultant fees, facility leases/usage fees, and/or other subcontract or purchase agreements, may not exceed one-half of the total contract price/cost, unless otherwise approved in writing by the government Contracting Officer. Provide copies of subcontract agreements. The supporting agreement documents must adequately describe the work to be performed. At a minimum, an applicant must include a Statement of Work (SOW) with a corresponding detailed Cost Volume (Volume 3) for each planned subcontract.
Copies of Consultant Agreements	Provide a separate agreement letter for each consultant. The letter should briefly state the tasks to be provided, hours required, and hourly rate. Some SBCs seek assistance from subcontractors or consultants to determine their solution's feasibility for a specific DoD customer. If working with a customer discovery firm like this, the associated agreement is required to be included in the proposal.
Facilities/Equipment	Describe instrumentation and/or physical/digital facilities necessary and available to carry out the proposed effort. Justify equipment to be purchased, including in 'Additional Cost Information,' as necessary. State whether proposed performance locations meet federal, state, and local government environmental laws and regulations for airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid/bulk waste disposal practices, toxic/hazardous material handling/storage, and others, as identified.
Indirect Costs	The proposal should identify the specific rates used and allocation bases to which they are applied. Provide proposed rates and applications per FY throughout the anticipated performance period. Do not propose composite rates.

Resumes of Key Personnel	Resumes of Key Personnel: Full resumes, including a publications list, for the Primary Investigator and other key technical personnel shall be included.
Non-Proprietary Work Plan	Provide a standalone, complete, non-proprietary work plan, free of proprietary markings. Failure to provide this document will necessitate a request for revision and may delay contract award. Note: this document is separate from the Work Plan Outline contained in Volume 2. The DAF often uses the Volume 5 version of the Work Plan as the initial draft of the Statement of Work (SOW) for the award.
DD Form 2345	International Traffic in Arms (ITAR) / Export Administration Regulations (EAR) Certification: For proposals whose solutions fall under export- controlled regulations (either International Traffic in Arms (ITAR) or EAR, a copy of the certified DD Form 2345, *Militarily Critical Technical Data Agreement*, or evidence of application submission must be included. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website, http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP/DD 2345Instructions.aspx. Approval of the DD Form 2345 will be verified if the proposal is chosen for award.
Allocation of Rights	If subcontracting to a research institution or other organization, the SBC may wish to complete an allocation of rights agreement. This is not required for SBIR awards but may be used at the company's discretion. Example: https://rt.cto.mil/wp-content/uploads/STTR-Model-Agreement-for-the-Allocation-of-Rights.pdf
Additional Letters of Support	Letters of support demonstrate that the proposed innovation, if developed, would solve a valid market need. More generally, letters of support can validate commercial impact claims made in the proposal. Therefore, letters from potential technology end users (e.g., customers and corporate partners/collaborators) are appropriate, as well as letters from actual or potential investors. All letters should be uploaded in a single .pdf file organized logically. Note: When combining .pdf documents, ensure digital signatures are not stripped.
Data Rights Assertions	If an applicant plans to submit assertions IAW DFARS 252.227-7017 or similar award term, those assertions must be identified, and restrictions MUST be included in the proposal submission in Volume 5. Award cannot be executed until assertions are approved or mutually agreed-upon during negotiations. Any SBIR/STTR data delivered by the Awardee, and in which the Awardee intends to limit the federal government's rights to SBIR/STTR data rights, must be delivered with restrictive markings. The federal government assumes no liability for the access, use, modification, reproduction, release, performance, display, disclosure, or distribution of SBIR/STTR data

	without markings. The Awardee or its subcontractors shall conspicuously and legibly mark all such SBIR/STTR data with the appropriate legend, placed on each page of the SBIR/STTR data.
Research Involving Human Subjects (RIHS)	Proposals that implicate RIHS must include a detailed description of the research plan, study population, risks, and benefits of study participation, recruitment and consent process, data collection and data analysis.

i. FRAUD, WASTE, AND ABUSE (FWA) TRAINING (VOLUME 6)

Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission. Additional details are addressed in the DoD SBIR Program CSO.

j. DISCLOSURES OF FOREIGN AFFILIATIONS OR RELATIONSHIPS TO FOREIGN COUNTRIES (VOLUME 7)

Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoD Program CSO.

k. MISCELLANEOUS

i. Common Disqualifiers

The following are common reasons for which proposals are disqualified. It is not a comprehensive list of potential reasons for disqualification but is meant to be instructive. Applicants shall read ALL solicitation instructions to ensure compliance.

- System for Award Management is not properly updated to include eligibility for ALL AWARDS.
- Government Engagement Memorandum missing required signature and/or content.
- Government Engagement Memorandum not signed by one of the mandatory Government customer organizations identified in the TRI.
- Application fails to reasonably demonstrate alignment with at least one of the stated manufacturing focus areas.
- Minimum Performance Percentage of Work is not allocated properly.
- Work, as proposed, does not meet the definition of R/R&D required for funding.
- Proposal submitted beyond deadline. In the event of technical difficulty, contact DSIP Help Desk immediately at DoDSBIRSupport@reisystems.com.
- Proposal submitted exceeds the maximum SBIR funding amount ($1,750,000.00)
- Proposal submitted exceeds the maximum specified duration (15 months).
- Proposal claims data rights assertions without including a *Data Rights Assertions* table (as found in DFARS 252.227-7017) in Volume 5, Supporting Documents, of the proposal.

ii. DAF SBIR Open Topic Objectives

The DAF SBIR Program's objectives include stimulating technological innovation, strengthening the small business role in meeting DoD research, development, test, and evaluation (RDT&E) needs,

fostering and encouraging small businesses in technological innovation, and increasing commercial application of DoD-supported RDT&E results.

The AF SBIR Program follows the policies and practices of the Small Business Administration (SBA) SBIR/STTR Policy Directive (3 May 2023): https://www.sbir.gov/sites/default/files/SBA%20SBIR_STTR_POLICY_DIRECTIVE_May2023.pdf.

iii. Proposal Review Information

The Air Force anticipates all proposal reviews will be complete, selection decisions made, and notification letters sent within approximately 90 calendar days of solicitation close. Please refrain from contacting the AF for proposal status before that time.

iv. Proposal Status and Feedback

The PI and Corporate Official indicated on the Proposal Coversheet will be notified by email regarding proposal selection or non-selection. The small business will receive one (1) notification for each proposal submitted. Please note the referenced proposal number and read each notification carefully. If changes occur to the company mail or email addresses or points of contact after proposal submission, the information must be provided to the DAF via AF SBIR/STTR One Help Desk at DoDSBIRSupport@reisystems.com.

Feedback will be provided to applicants with proposals designated as "Not Selected". The feedback provided with the "Not Selected" notice will constitute the only feedback applicants will receive. Requests for additional feedback will not receive a response. The DAF anticipates fulfilling feedback requests using automated standardized language associated with peer review/technical evaluation scores. As such, applicants will not be provided with selection decision documentation nor will feedback documents constitute selection documentation. Note: Feedback is not the same as a FAR Part 15 debriefing.

Acquisitions under this solicitation are awarded via "other competitive procedures" IAW DFARS 206.102-70. Therefore, applicants are neither entitled to nor will they be provided FAR Part 15 debriefs.

v. Use of Support Contractors for the DAF SBIR Program

Proposals may be handled for administrative purposes only by support contractors, including APEX, Oasis Systems, Riverside Research, Peerless Technologies, HPC- COM, Mile Two, MTSI, Reef Systems, Ernst & Young, Wright Brothers Institute, Parallax, TEC Solutions, Inc. and MacB (an Alion Company). In addition, only Government employees and technical personnel from Federally Funded Research and Development Centers (FFRDCs) MITRE and Aerospace Corporation working under contract to provide technical support to AF Life Cycle Management Center and Space and Missiles Centers may evaluate proposals. All support contractors are bound by appropriate non-disclosure agreements. Contact the Air Force SBIR/STTR Contracting Officer (Daniel.Brewer.13@us.af.mil) with concerns regarding the listed contractors.

l. MANUFACTURING CHALLENGE TOPIC AFX255-DPCSO6

AFX255-DPCSO6 TITLE: Direct-to-Phase-II Call for Innovative Defense-Related Dual-Purpose Manufacturing Technologies/Solutions with a Clear Air Force and Space Force Stakeholder Need

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Trusted AI and Autonomy; Directed Energy (DE); Hypersonics; Microelectronics; Integrated Network Systems-of-Systems; Quantum Science; Space Technology; Advanced Materials; Human-Machine Interfaces; Advanced Infrastructure & Advanced Manufacturing

OBJECTIVE: This focused open topic is specific to **manufacturing**. Applications must reasonably demonstrate the proposed solution falls within at least one or more of the following focus areas: engine production with advanced manufacturing, advanced composite structures, semiconductor production innovations, radiation-tolerant/hardened electronics, system integration and interoperability, and affordable energetics.

This Department of the Air Force (DAF) focused open topic seeks Manufacturing Technologies for transition into the United States Air Force (USAF) and United States Space Force (USSF). This topic is offered in partnership with AFWERX/SpaceWERX.

Primary objectives of this focused open topic include exploring innovative manufacturing solutions applicable to both defense and non-defense markets, scaling capability, and growing the industrial base for defense. This topic seeks companies with the capability to prototype validated concepts under an accelerated Phase II schedule. This topic is aimed at applied research and development efforts rather than "front-end" or basic research (R)/Research & Development (R&D).

Topic Unique Government Engagement Memorandum (GEM) Requirements

a. Your application *must* include a Government Engagement Memorandum (Attachment 1) that proves you have support from a valid DAF *customer* within one of the following organizations: **PEO Weapons/Stand-in Strike Division, PEO Fighters and Advanced Aircraft/Advanced Aircraft Division, AFRL/RX, or AFRL/RG Prime.**

***A valid signature from one of the listed Customer organizations *must* be present on the Government Engagement Memorandum for your application to be considered. Applications lacking a completed Government Engagement Memorandum will be disqualified.** The Government Engagement Memorandum is the only document accepted to verify DAF customer support.

b. Applicants ARE NOT required to obtain TPOC and End-User signatures. TPOCs and End-Users will be identified by the customer if the proposal is selected for award.

c. To aid applicants' connection with potential DAF customers an optional process is being offered to facilitate communication. This process allows small businesses to share information about their technologies, allowing government partners to determine if they have an interest in providing a Government Engagement Memorandum of support via completion of an online webform.

d. Webform Process

i. Applicants choosing to engage in this optional process will complete a web form

following this link https://forms.gle/A1zinCu6Q9xqs1ji7. Applicants engaging in this process shall submit the completed web form not later than **12:00 PM ET on 18 June 2025**. After this time the webform will close to new submissions to allow potential Government customers a reasonable time to review applicant submissions.

Upon submission of the requested information via the webform, Government customers will review it to determine their interest in the technology.

If any Government customer(s) have an interest in providing a Customer Engagement Memorandum to the applicant, the Government customer(s) will communicate their intent to sign the Government Engagement Memorandum. Further, the Government customer, by signature and completion of the Government Engagement Memorandum, commits that the Government signatory will identify by name a valid TPOC and End-User **not later than September 15th, 2025** or at another time as agreed in writing by an AFWERX Contracting Officer.

Up to 20 applicants can receive a signed Government Engagement Memo from each of the four specified Government customer organizations.

ii. The webform will request the following information from applicants:

- Name of company
- Company POC name/contact information
- Alternate POC name/contact information
- Technical abstract (no more than 500 words)
- Anticipated benefits of the research or development toward the six identified Manufacturing focus areas(no more than 300 words)
- Identification of one or more of the identified customer organizations the company believes may be interested in their proposed solution

e. The DAF prefers applicants use the webform to streamline the process. However, applicants, at their sole discretion, may choose to source their Government Engagement Memorandum signatures directly with the four identified Government customer organizations.

f. Customer Organization Missions and Points of Contact

i. PEO Weapons/Armament Directorate/Stand-in Strike Division (AFLCMC/EBI): Designs, develops, produces, fields, and sustains a family of air-to-ground and air-to-air munitions for both U.S. and allied nations to defeat a spectrum of enemy targets. AFRL.RG.PEOWeapons@us.af.mil

ii. PEO Fighters and Advanced Aircraft/Advanced Aircraft Division (AFLCMC/WAC): Responsible for ensuring the U.S. Air Force and its allies have the most effective fighter and advanced aircraft. The directorate manages everything from the initial development and production to the long-term maintenance and upgrades of aircraft like the A-10, F-15, F-16, and F-22, as well as special programs. Fighters and Advanced Aircraft (AFLCMC/WAC) is also spearheading the development of the future of air dominance with the Next Generation Air Dominance (NGAD) program. The PEO's focus is on constantly delivering cutting-edge capabilities through innovative strategies to stay ahead of potential threats. AFRL.RG.PEOFighters@us.af.mil

iii. AFRL's Materials and Manufacturing Directorate (AFRL/RX): Develops materials, processes, and advanced manufacturing technologies for aircraft, spacecraft, missiles, rockets, and ground-based systems and their structural, electronic and optical components. Air Force product centers,

logistic centers, and operating commands rely on the directorate's expertise in materials, nondestructive inspection, systems support, and advanced manufacturing methods to solve system, expeditionary deployment, and operational challenges. AFRL.RG.Materials@us.af.mil

iv. AFWERX Prime (AFRL/RGP): Cultivates innovative technologies and empowers Airman and Guardian talent to mature groundbreaking capabilities for rapid operational impact. AFRL.RG.AFWERXPrime@us.af.mil

Manufacturing Technology Focus Areas and Definitions

Proposed solutions for this **Manufacturing Technology** focused open topic must fall within at least one of the following six (6) focus areas:

- **Engine Production with Advanced Manufacturing:** Innovations in manufacturing processes, materials, and automation to improve the performance, efficiency, cost basis, and reliability of engines for Air Force and Space Force applications. This includes, but is not limited to, additive manufacturing, digital twins, and advanced robotics for engine component production and assembly for aircraft, spacecraft, other flying systems, incremental sheet forming for engine structures (ducts, cases, etc.), oxide dispersion stabilized materials for hot engine components, environmental coatings opportunities, and other relevant technologies.

- **Advanced Composite Structures:** Development and application of novel materials, manufacturing techniques, and structural designs for high-performance composite materials for Air Force and Space Force applications. This includes, but is not limited to, automation of composite manufacturing, advanced fiber placement, and innovative joining methods for aerospace vehicles, space structures and related equipment, graphene-based materials, integrated isogrids to reduce part count, high pressure molding to reduce production cycle times, utilization of chopped fiber to create complex parts at high production rates, low cost Thermoset (TS) and Thermoplastic (TP), Thermoplastic Welding and weld inspections, automation solutions for final stage assembly, and other relevant technologies.

- **Semiconductor Production Innovations:** Advancements in semiconductor manufacturing processes, equipment, and materials to improve device performance, reduce costs, and increase production efficiency for Air Force and Space Force systems. This includes, but is not limited to, innovations in lithography, etching, deposition, and packaging technologies for radiation-hardened electronics, advanced sensors, communication systems, nano imprint lithography, wide-band gap semiconductors, lightweight semiconductors, 3D advance packaging (PCB board elimination), and other relevant technologies.

- **Radiation-Tolerant/Hardened Electronics:** Development of manufacturing techniques and materials for electronic components and systems that can withstand harsh radiation environments relevant to Air Force and Space Force missions. This is critical for applications in space-based systems, high-altitude platforms, hypersonics, and nuclear environments.

- **System Integration and Interoperability:** Innovations in processes and technologies to improve integration and interoperability of defense relevant manufacturing equipment and systems. This includes, but is not limited to, digital manufacturing, manufacturing data management and utilization, manufacturing system integration tools and advanced assembly techniques for vehicles, platforms, components of air and space systems, and other relevant technologies.

- **<u>Affordable Energetics:</u>** Development of cost-effective manufacturing methods for energetic materials and devices, including propellants, explosives, and pyrotechnics, and their containers or encasements for Air Force and Space Force applications. This includes, but is not limited to, innovations in synthesis, processing, and formulation of energetic materials for propulsion systems, and novel, low-cost materials and manufacturing techniques for energetics cases or containers, for missile systems, high-gurney self-sustaining energetics, and other relevant technologies.

Applications for this Manufacturing Technologies focused open topic must clearly demonstrate that the proposed solution fits within one of the six (6) defined focus areas. Applications failing to reasonably validate their alignment within one or more of the focus areas will not be evaluated.

DESCRIPTION: The DAF is a large and complex organization consisting of many functions, a vast majority of which have similar counterparts in the commercial sector. It is important that potential solutions have a high probability of keeping pace with technological change. Thus, solutions should be closely tied to commercial technologies and solutions supporting the proposed development. This topic is meant for non-defense commercial solutions to be adapted to meet DAF stakeholders' needs.

Proposed technical solutions should focus on the aforementioned 6 focus areas as well as the characteristics outlined in the Selection and Evaluation Criteria section of this solicitation.

This is accomplished most effectively through a proposal with a mature non-defense technical solution and a clear understanding of its adaptation to meet a DAF customer's specific need, supported by documentation from a specific motivated, empowered DAF customer ready and willing to participate in the proposed prototype solution's trial.

The AF SBIR Program follows the policies and practices of the Small Business Administration (SBA) SBIR/STTR Policy Directive (3 May 2023):
https://www.sbir.gov/sites/default/files/SBA%20SBIR_STTR_POLICY_DIRECTIVE_May2023.pdf.

PHASE I: This topic is intended for technology proven ready to move directly into Phase II. Therefore, Phase I awards will not be made for this topic. The applicant is required to provide detail and documentation in the D2P2 proposal which demonstrates accomplishment of a “Phase I-type” effort, including a feasibility study. This includes determining, insofar as possible, the scientific and technical merit and feasibility of ideas appearing to have commercial potential. It must have validated the product-mission fit between the proposed solution and a potential Air Force stakeholder. The applicant should have defined a clear, immediately actionable plan with the proposed solution and the DAF customer.

The feasibility study should have:
1. Clearly identified the potential stakeholders of the adapted solution for solving the Air Force need(s).
2. Described the pathway to integrating with DAF operations, to include how the applicant plans to accomplish core technology development, navigate applicable regulatory processes, and integrate with other relevant systems and/or processes.
3. Described if and how the solution can be used by other Department of Defense (DoD) or governmental customers.

PHASE II: Proposals should include development, installation, integration, demonstration and/or test and evaluation of the proposed solution prototype system.

These activities should focus specifically on:
1. Evaluating the adapted solution against the proposed objectives and measurable key results.
2. Describing in detail how the installed solution differs from the non-defense commercial offering to solve the Air Force need(s), as well as how it can be scaled for wide adoption, i.e., modified for scale.
3. Identifying the proposed solution's clear transition path, taking into account input from affected stakeholders, including but not limited to, end users, engineering, sustainment, contracting, finance, legal, and cyber security.
4. Specifying the solution's integration with other current and potential future solutions.
5. Describing the solution's sustainability, i.e., supportability. Identifying other specific DoD or governmental customers for the solution.

PHASE III DUAL-USE APPLICATIONS: Some solutions may go from Phase II to Phase III as soon as the product-market fit is verified. Potential Phase III awardees will transition to the adapted non-Defense commercial solution to provide expanded mission capability for a broad range of potential governmental and civilian users and alternate mission applications.

NOTES: DAF may hold Ask Me Anything (AMA) webinars to answer applicant questions. For more information, applicants may visit https://afwerxchallenge.com/.

AMAs and webinars will be held at the discretion of the DAF and are subject to change.

Section seven (7) of the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Extension Act of 2022 requires the DoD to establish innovation open topic activities in order to:

(A) Increase the transition of commercial technology to the DoD;
(B) Expand the small business non-traditional industrial base;
(C) Increase commercialization derived from investments of the DoD; and
(D) Expand the ability for qualifying small business concerns to propose technology solutions to meet the needs of the DoD.

Unlike conventional topics, which specify the desired technical objective and output, open topics can use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

REFERENCES:
1. United States Department of Defense. (2023). National Defense Science & Technology Strategy 2023.
2. United States Department of Defense. (2022). 2022 National Defense Strategy of the United States Summary.
3. United States Space Force. (2024). Commander’s Strategic Guidance.
4. National Science and Technology Council. (2018). Strategy For American Leadership In Advanced Manufacturing.
5. United States Department of Defense. (2021). Department of Defense Additive Manufacturing Strategy
6. Executive Order No. 13329 (2004). Encouraging Innovation in Manufacturing.

KEY WORDS: Open; Other; Disruptive; Radical; Dual-Use; Commercial; Additive Manufacturing; Advanced Manufacturing; Smart Manufacturing; Industry 4.0; Digital Thread; Digital Twin; Advanced Materials; Robotics

m. ATTACHMENTS

Attachment 1: Government Engagement Memorandum (required)

The Government Engagement Memorandum is found at https://afwerx.com/divisions/ventures/open-topic/.

The correct Government Engagement Memorandum for this solicitation will be labeled "25.5 Release 9 D2P2" at the top of the memorandum. It is the only accepted version.

Note 1: Applicants must use the version of the fillable template linked in this solicitation to fulfill this requirement. Proposals that source Government Engagement Memoranda elsewhere **will be disqualified**.

Note 2: Proposals including documents purporting to be Government Engagement Memorandum that are not completed using the mandatory template **will be disqualified.**

Note 3: Documents that are substantively or substantially similar and/or identical to the Government Engagement Memorandum, but authored without the use of the mandatory template, will not be accepted by the Government and proposals containing such documents **will be disqualified.**

Note 5: Applications lacking a signed Government Engagement Memorandum showing support from an authorized customer within PEO Weapons/Stand-in Strike Division, PEO Fighters and Advanced Aircraft/Advanced Aircraft Division, AFRL/RX, or AFRL/RG Prime, will be disqualified..

Attachment 2: Regulatory Compliance Initial Safety Assessment Form (required)

The Regulatory Compliance Initial Safety Assessment Form is found at https://afwerx.com/divisions/ventures/open-topic/.

Attachment 3: SBIR/STTR Funding Agreement Certification (required prior to award)

Attachment 4: SBIR/STTR Funding Agreement Certification—Life Cycle Certification (required prior to award)

Attachment 5: Phase II Funding Commitment Form (required if applicant proposes matching non-SBIR governmental funding or matching private funding)

Attachment 6: Identification and Assertion of Use, Release, or Disclosure Restrictions (required if applicable)

Attachment 7: Certification for SBIR Applicants that are Majority-Owned by Multiple Venture Capital Operating Companies, Hedge Funds, or Private Equity Firms (required if applicable)

ATTACHMENT 3 (required)
SBIR/STTR FUNDING AGREEMENT CERTIFICATION

All small businesses that are selected for award of an SBIR/STTR Funding Agreement must complete this certification at the time of award and any other time set forth in the Funding Agreement that is prior to performance of work under this award. This includes checking all of the boxes and having an authorized officer of the Awardee sign and date the certification each time it is requested.

Please read carefully the following certification statements. The Federal Government relies on the information to determine whether the business is eligible for a Small Business Innovation Research (SBIR) program or Small Business Technology Transfer (STTR) program award. A similar certification will be used to ensure continued compliance with specific program requirements during the life of the Funding Agreement. The definitions for the terms used in this certification are set forth in the Small Business Act, SBA regulations (13 CFR part 121), the SBIR/STTR Policy Directive and also any statutory and regulatory provisions referenced in those authorities.

If the Funding Agreement officer believes that the business may not meet certain eligibility requirements at the time of award, they are required to file a size protest with the U.S. Small Business Administration (SBA), which will determine eligibility. At that time, SBA will request further clarification and supporting documentation in order to assist in the verification of any of the information provided as part of a protest. If the Funding Agreement officer believes, after award, that the business is not meeting certain Funding Agreement requirements, the agency may request further clarification and supporting documentation in order to assist in the verification of any of the information provided.

Even if correct information has been included in other materials submitted to the Federal Government, any action taken with respect to this certification does not affect the Government's right to pursue criminal, civil or administrative remedies for incorrect or incomplete information given in the certification. Each person signing this certification may be prosecuted if they have provided false information.

The undersigned has reviewed, verified and certifies that (all boxes must be checked unless otherwise directed):

☐ (1) The Awardee business concern meets the ownership and control requirements set forth in 13 CFR 121.702.

(2) If a corporation – all corporate documents(namely: articles of incorporation and any amendments, articles of conversion, by-laws and amendments, shareholder meeting minutes showing director elections, shareholder meeting minutes showing officer elections, organizational meeting minutes, all issued stock certificates, stock ledger, buysell agreements, stock transfer agreements, voting agreements, and documents relating to stock options, including the right to convert non-voting stock or debentures into voting stock) must evidence that the corporation meets the ownership and control requirements set forth in 13 CFR 121.702. (Check one box).

☐ Yes

☐ N/A (Explain why N/A)

(3) If a partnership -- the partnership agreement evidences that it meets the ownership and control requirements set forth in 13 CFR 121.702. (Check one box).

☐ Yes

☐ N/A (Explain why N/A)

(4) If a limited liability company – the articles of organization and any amendments, and operating agreement and amendments, evidence that it meets the ownership and control requirements set forth in 13 CFR 121.702. (Check one box).

☐ Yes

☐ N/A (Explain why N/A)

(5) The birth certificates, naturalization papers, or passports show that any individuals it relies upon to meet the eligibility requirements are U.S. citizens or permanent resident aliens in the United States. (Check one box).

☐ Yes

☐ N/A (Explain why N/A)

☐ (6) The Awardee business concern has no more than 500 employees, including the employees of its Affiliates.

☐ (7) SBA has not issued a size determination currently in effect finding that this business concern exceeds the 500 employee size standard.

(8) During the performance of the award, the Principal Investigator/Project Manager will spend more than one half of his/her time (based on a 40 hour workweek) as an employee of the Awardee (or Research Institution – STTR only) or has requested and received a written deviation from this requirement from the Funding Agreement officer. (Check one box).

☐ Yes

☐ Deviation requested in writing to the Funding Agreement officer: _____%

(9) All Essentially Equivalent Work, or a portion of the work, proposed under this project (check applicable line):

☐ **Has not** been submitted for funding to this Agency or another Federal agency.

☐ **Has** been submitted for funding to this Agency or another Federal agency but has not been funded under any other grant, contract, subcontract or other transaction.

☐ A portion has been funded by another grant, contract, or subcontract as described in detail in the proposal and approved in writing by the Funding Agreement officer.

(10) During performance of award, the Awardee will perform the applicable percentage of work unless a deviation from this requirement is approved in writing by the Funding Agreement officer (check applicable line and fill in if needed):

☐ SBIR Phase I: at least two-thirds (66 2/3%) of the research.

☐ SBIR Phase II: at least half (50%) of the research.

☐ STTR Phase I or Phase II: at least forty percent (40%) of the research. Deviation approved in writing by the Funding Agreement officer (SBIR only): __%

(11) During performance of award, the R/R&D will be performed in the United States unless a deviation is approved in writing by the Funding Agreement officer (check one box).

☐ Yes

☐ Waiver has been requested

☐ (12) During performance of award, the R/R&D will be performed at the Awardee's facilities by the Awardee's employees, except as otherwise indicated in the SBIR/STTR application and approved in the Funding Agreement.

(13) The SBIR Awardee has registered itself on SBA's database as majority-owned by venture capital operating companies, hedge funds or private equity firms (check one)

☐ Yes

☐ No

☐ N/A Explain why N/A: _____

(14) The SBIR Awardee is a Covered Small Business Concern (a Small Business Concern that: (a) was not majority-owned by multiple venture capital operating companies (VCOCs), hedge funds, or private equity firms on the date on which it submitted an application in response to an SBIR solicitation; and (b) on the date of the SBIR award, which is made more than 9 months after the closing date of the solicitation, is majority-owned by multiple venture capital operating companies, hedge funds, or private equity firms). (Check one box).

☐ Yes

☐ No

☐ (15) I will notify this Agency immediately if all or a portion of the work authorized and funded under this award is subsequently funded by another Federal Agency.

(16) [For STTR only] The Small Business Concern, and not a partnering Research Institution, is exercising management direction and control of the performance of the STTR Funding Agreement.

☐ Yes

☐ No

☐ (17) I understand that the information submitted may be given to Federal, State, and local agencies for determining violations of law and other purposes.

☐ (18) I am an officer of the business concern authorized to represent it and sign this certification on its behalf. By signing this certification, I am representing on my own behalf, and on behalf of the business concern that the information provided in this certification, the application, and all other information submitted in connection with this application, is true and correct as of the date of submission. I acknowledge that any intentional or negligent misrepresentation of the information contained in this certification may result in criminal, civil or administrative sanctions, including but not limited to: (1) fines, restitution and/or imprisonment under 18 U.S.C. 1001; (2) treble damages and civil penalties under the False Claims Act (31 U.S.C. 3729 et seq.); (3) double damages and civil penalties under the Program Fraud Civil Remedies Act (31 U.S.C. 3801 et seq.); (4) civil recovery of award funds, (5) suspension and/or debarment from all Federal procurement and nonprocurement transactions (FAR subpart 9.4 or 2 CFR part 180); and (6) other administrative penalties including termination of SBIR/STTR awards.

Signature	
Print Name	
Title	
Business Name	

ATTACHMENT 4
SBIR/STTR FUNDING AGREEMENT CERTIFICATION – LIFE CYCLE CERTIFICATION (Required)

All SBIR/STTR Phase I and Phase II Awardees must complete this certification at all times set forth in the Funding Agreement (see § 8(j) of the SBIR/STTR Policy Directive). This includes checking all of the boxes (unless otherwise directed) and having an authorized officer of the Awardee sign and date the certification each time it is requested.

If the Funding Agreement officer believes that the business is not meeting certain Funding Agreement requirements, the agency may request further clarification and supporting documentation in order to assist in the verification of any of the information provided.

Even if correct information has been included in other materials submitted to the Federal Government, any action taken with respect to this certification does not affect the Government's right to pursue criminal, civil or administrative remedies for incorrect or incomplete information given in the certification. Each person signing this certification may be prosecuted if they have provided false information.

The undersigned has reviewed, verified and certifies that (all boxes must be checked except where otherwise directed):

(1) The Principal Investigator/Project Manager spent more than one half of his/her time (based on a 40 hour workweek) as an employee of the Awardee (or Research Institution – STTR only) or the Awardee has requested and received a written deviation from this requirement from the Funding Agreement officer.

☐ Yes

☐ No

☐ Deviation approved in writing by Funding Agreement officer: _____%

(2) All Essentially Equivalent Work, or a portion of the work, performed under this project (check the applicable line):

☐ Has not been submitted for funding to this Agency or another Federal Agency.

☐ Has been submitted for funding to this Agency or another Federal agency but has not been funded under any other grant, contract, subcontract or other transaction.

☐ A portion has been funded by another grant, contract, or subcontract as described in detail in the proposal and approved in writing by the Funding Agreement officer.

(3) Upon completion of the award, the Awardee will have performed the applicable percentage of work, unless a deviation from this requirement is approved in writing by the Funding Agreement officer (check the applicable line and fill in if needed):

☐ SBIR Phase I: at least two-thirds (66 2/3%) of the research.

☐ SBIR Phase II: at least half (50%) of the research.

☐ STTR Phase I or Phase II: at least forty percent (40%) of the research. Deviation approved in writing by the Funding Agreement officer (SBIR only): _____%

☐ (4) The work is completed and the small business Awardee has performed the applicable percentage of work, unless a deviation from this requirement is approved in writing by the Funding Agreement officer (check the applicable line and fill in if needed):

☐ SBIR Phase I: at least two-thirds (66 2/3%) of the research.

☐ SBIR Phase II: at least half (50%) of the research.

☐ STTR Phase I or Phase II: at least forty percent (40%) of the research. Deviation approved in writing by the Funding Agreement officer: N/A because work is not completed. ☐

(5) [For STTR only] The Small Business Concern, and not a partnering Research Institution, is exercising management direction and control of the performance of the STTR Funding Agreement.

☐ Yes

☐ No

(6) The R/R&D is performed in the United States unless a deviation is approved in writing by the Funding Agreement officer.

☐ Yes

☐ No

☐ Waiver has been granted

(7) The R/R&D is performed at the Awardee's facilities by the Awardee's employees, except as otherwise indicated in the SBIR/STTR application and approved in the Funding Agreement.

☐ Yes

☐ No

☐ (8) I will notify this Agency immediately if all or a portion of the work authorized and funded under this award is subsequently funded by another Federal Agency.

☐ (9) I understand that the information submitted may be given to Federal, State, and local agencies for determining violations of law and other purposes.

☐ (10) I am an officer of the Awardee business concern authorized to represent it and sign this certification on its behalf. By signing this certification, I am representing on my own behalf, and on behalf of the business concern, that the information provided in this certification, the application, and all other information submitted in connection with the award, is true and correct as of the date of submission. I acknowledge that any intentional or negligent misrepresentation of the information contained in this certification may result in criminal, civil or administrative sanctions, including but not limited to: (1) fines, restitution and/or imprisonment under 18 U.S.C. 1001; (2 124 damages and civil penalties under the False Claims Act (31 U.S.C. 3729 et seq.); (3) double damages and civil penalties under the Program Fraud Civil Remedies Act (31 U.S.C. 3801 et seq.); (4) civil recovery of award funds, (5) suspension and/or debarment from all Federal procurement and nonprocurement transactions (FAR subpart 9.4 or 2 CFR part 180); and (6) other administrative penalties including termination of SBIR/STTR awards.

Signature	
Print Name	
Title	
Business Name	

ATTACHMENT 5
PHASE II FUNDING COMMITMENT FORM (Required if applicable)

1. For Governmental (other than SBIR) funding:

Government Funding Organization	Government POC Name/E-mail	Amount of Funding	Year/Type Funding, e.g., RDT&E, O&M, etc.

2. For private investment:

Private Organization Funding	Private POC Name/E-mail	Amount of Funding	Type of Funding, e.g., pre-sales, etc.

3. Total Government (other than SBIR) Funding Amount: $_____

4. Total Private Funding Amount: $_____

5. Total Requested SBIR Funding Amount: $_____

6. Government (other than SBIR) Funding Work Plan Summary:

[INSERT brief statement describing the project portion the Government sponsor will fund, i.e., how does it support the Phase II Work Plan, additional R&D, or marketing under the proposed SBIR project?]

7. Private Funding Work Plan Summary:

[INSERT brief statement describing the project portion the investor will fund, i.e., how does it support the Phase II Work Plan, additional R&D, or marketing under the proposed SBIR project?]

Voluntary Phase II Funding Commitment Guidance

i. Examples of Voluntary Phase II Funding Commitments

Investment from outside sources, which includes other companies, state or local Governments, venture capital firms, individual "angel" investors, non-SBIR/non-STTR Federal Government programs, or any combination thereof, is considered such a commitment. This does not include the small business's owner, family members, and/or "affiliates" of the small business, as defined in 13 CFR Section 121.103, as highlighted below:

ii. Affiliates

Concerns are affiliates when one concern controls or has the power to control the other, or a third party or parties controls or has the power to control both.

iii. Affiliation Determinations

Factors such as ownership, management, previous relationships with or ties to another concern, and contractual relationships are considered in making affiliation determinations.

iv. Treatment as One Party

Individuals or firms with identical or substantially identical business or economic interests, such as family members, persons with common investments, or firms economically dependent through contractual or other relationships, may be treated as one party with such interests aggregated. While the proposal must state whether funding is from external or internal sources, the CO and the SBA have the discretion to designate an entity as an "outside investor," regardless of the designation in the proposal.

v. Commitment Arrangements

Such a commitment may also be an arrangement by which the outside party provides cash to the small company in return for such items as equity; a share of royalties; rights in the technology; a percentage of profit; an advance purchase order for products resulting from the technology; Federal funds transferred to AFWERX Financial Management (AFRL/RGF) via Military Interdepartmental Purchase Request (MIPR); or any combination thereof.

vi. The DAF reserves the right to adjust these guidelines at any time. Any changes to these guidelines will be sent to the eligible companies and posted online via amendment to this solicitation.

vii. ***Can Entity E's investment in Small Company S during the first month of S's Phase I SBIR performance qualify as a second state commitment?***
Yes, provided E is an outside investor and the other Voluntary Second Stage Commitment conditions are met. The investment can occur at any time after the start of Phase I performance.

viii. ***Small Company A, which won a Phase I award, spins off Small Company B to commercialize the SBIR technology. A then convinces Angel Investor I to invest funds in B. Is I's investment in B considered a second stage commitment?***
For Angel Investor I's investment in Small Company B to qualify Small Company A for such a commitment, AF must determine if A and B are substantially the same entity as evidenced, for example, by meeting the "affiliates" definition " in 13 C.F.R. Sec.121.103. If AF determines A and B are substantially the same entity, I's investment in B could qualify A as a second stage commitment. Of course, the parties must also meet the other conditions, e.g., I must be an outside investor.

ix. ***Can Entity E's investment in Small Company S before the submission of S's Direct to Phase II SBIR proposal qualify as a Phase II funding commitment?***
Yes, provided E is an outside investor and the other second stage commitment conditions are met. The investment can occur at any time after the release date of Phase II proposal instructions, and no later than 90 calendar days after award.

Applicants wishing to include Phase II funding commitments, Government or private, in their proposals must submit the information via a completed mandatory Phase II Funding Commitment form, as described in this section.

For Governmental second phase commitments that are to be applied directly to the contract, DRAFT funding documents (Form 9s/MIPRs) are due within 45 days of the solicitation closing. Once selection letters are sent, CERTIFIED funding documents are due five business days later. If certified funding documents are not provided, the effort will move forward without the second phase funding commitment considered. To initiate governmental funding transfer, the interested government organization should send an email to the AFRL/RGF Incoming Documents mailbox, AFRL.SBIR.STTR@us.af.mil. The sender will receive an auto-response with guidance regarding incoming funds submission and associated information required. RGF requests a draft MIPR/Form 9 be submitted to the same inbox before certification to ensure the document is complete and accurate.

NOTE 1: If sending funding via MIPR, RGF requires a signed MOA with the submitting Government organization.

NOTE 2: If providing other than RDT&E 3600 funds, the corresponding proposal milestones must clearly delineate deliverables for which the non-RDT&E funds will be used. Additionally, the submitting and receiving Comptrollers must agree to the funds propriety for the specified deliverables.

*The government organization must initiate communication with AFRL/RGF, not the small business.

ATTACHMENT 6
IDENTIFICATION AND ASSERTION OF USE, RELEASE, OR DISCLOSURE RESTRICTIONS
(required if applicable)

Technical Data or Computer Software to be Furnished with Restrictions*	Basis for Assertion**	Asserted Rights Category***	Name of Person Asserting Restrictions

*For technical data (other than computer software documentation) pertaining to items, components, or processes developed at private expense, identify both the deliverable technical data and each such item, component, or process. For computer software or computer software documentation identify the software or documentation.
** Generally, development at private expense, either exclusively or partially, is the only basis for asserting restrictions. For technical data, other than computer software documentation, development refers to development of the item, component, or process to which the data pertain. The Government's rights in computer software documentation generally may not be restricted. For computer software, development refers to the software. Indicate whether development was accomplished exclusively or partially at private expense. If development was not accomplished at private expense, or for computer software documentation, enter the specific basis for asserting restrictions.
*** Enter asserted rights category (e.g., government purpose license rights from a prior contract, rights in SBIR data generated under another contract, limited, restricted, or government purpose rights under this or a prior contract, or specially negotiated licenses).
**** Corporation, individual, or other person, as appropriate.

Signature	
Print Name	
Title	

***** Under the first column, only identify technical data and computer software/documentation that will be **delivered** with restrictions. Further, for each technical data item identified in the first column, include the applicable CLIN that it will be delivered under (e.g. CLIN 0002). For **deliverable** non-commercial software, identify the applicable CLIN (e.g. CLIN 0001).)

ATTACHMENT 7
CERTIFICATION FOR SBIR APPLICANTS THAT ARE MAJORITY-OWNED BY MULTIPLE VENTURE CAPITAL OPERATING COMPANIES, HEDGE FUNDS, OR PRIVATE EQUITY FIRMS (Required if applicable)

Any small business that is majority-owned by multiple venture operating companies (VCOCs), hedge funds, or private equity firms and is submitting an application for an SBIR Funding Agreement must complete this certification prior to submitting an application. This includes checking all of the boxes and having an authorized officer of the Applicant sign and date the certification each time it is requested.

Please read carefully the following certification statements. The Federal Government relies on the information to determine whether the business is eligible for a Small Business Innovation Research (SBIR) program award and meets the specific program requirements during the life of the Funding Agreement. The definitions for the terms used in this certification are set forth in the Small Business Act, SBA regulations (13 CFR part 121), the SBIR/STTR Policy Directive and also any statutory and regulatory provisions referenced in those authorities.

If the Funding Agreement officer believes that the business may not meet certain eligibility requirements at the time of award, he/she is required to file a size protest with the U.S. Small Business Administration (SBA), which will determine eligibility. At that time, SBA will request further clarification and supporting documentation in order to assist in the verification of any of the information provided as part of a protest. If the Funding Agreement officer believes, after award, that the business is not meeting certain Funding Agreement requirements, the agency may request further clarification and supporting documentation in order to assist in the verification of any of the information provided.

Even if correct information has been included in other materials submitted to the Federal Government, any action taken with respect to this certification does not affect the Government's right to pursue criminal, civil or administrative remedies for incorrect or incomplete information given in the certification. Each person signing this certification may be prosecuted if they have provided false information.

The undersigned has reviewed, verified and certifies that (all boxes must be checked):

(1) ☐ The Applicant is NOT more than 50% owned by a single VCOC, hedge fund, or private equity firm.

(2) ☐ The Applicant is more than 50% owned by multiple domestic business concerns that are VCOCs, hedge funds, or private equity firms.

(3) ☐ I have registered with SBA at www.SBIR.gov as a business that is majority-owned by multiple VCOCs, hedge funds or private equity firms.

(4) ☐ I understand that the information submitted may be given to Federal, State, and local agencies for determining violations of law and other purposes.

(5) ☐ All the statements and information provided in this form and any documents submitted are true, accurate, and complete. If assistance was obtained in completing this form and the supporting documentation, I have personally reviewed the information and it is true and accurate. I understand that, in general, these statements are made for the purpose of determining eligibility for an SBIR Funding Agreement and continuing eligibility.

(6) ☐ I understand that the certifications in this document are continuing in nature. Each SBIR Funding Agreement for which the small business submits an offer or application or receives an award constitutes a restatement and reaffirmation of these certifications.

(7) ☐ I understand that I may not misrepresent status as small business obtain a contract under the Act; or 2) obtain any benefit under a provision of Federal law that references the SBIR program.

(8) ☐ I am an officer of the business concern authorized to represent it and sign this certification on its behalf. By signing this certification, I am representing on my own behalf, and on behalf of the SBIR Applicant or Awardee, that the information provided in this certification, the application, and all other information submitted in connection with this application, is true and correct as of the date of submission. I acknowledge that any intentional or negligent misrepresentation of the information contained in this certification may result in criminal, civil or administrative sanctions, including but not limited to: (1) fines, restitution and/or imprisonment under 18 U.S.C. 1001; (2) treble damages and civil penalties under the False Claims Act (31 U.S.C. 3729 et seq.); (3) double damages and civil penalties under the Program Fraud Civil Remedies Act (31 U.S.C. 3801 et seq.); (4) civil recovery of award funds, (5) suspension and/ or debarment from all Federal procurement and nonprocurement transactions (FAR subpart 9.4 or 2 CFR part 180); and (6) other administrative penalties including termination of SBIR/STTR awards

Signature	
Print Name	
Title	
Business Name	